

# CareConnect

## Technical Design Document

University of Maryland Global Campus

Software Engineering 670 9040 Software Engineering Capstone

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## Revision History

Revision History Table

| Team Name | Date       | Reason for changes          | Version |
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# Table of Contents

|                                                      |          |
|------------------------------------------------------|----------|
| <b>Revision History</b>                              | <b>2</b> |
| Revision History Table                               | 2        |
| Table of Contents                                    | 3        |
| 1. Introduction                                      | 6        |
| 1.1 Purpose                                          | 6        |
| 1.2 Intended Audience                                | 6        |
| 1.3 Document Scope                                   | 6        |
| Included in Scope                                    | 6        |
| Excluded from Scope                                  | 6        |
| 1.4 Definitions, Acronyms, and Abbreviations         | 7        |
| 1.5 References                                       | 7        |
| 2. System Overview                                   | 8        |
| 2.1 System Description                               | 8        |
| 2.2 System Objectives                                | 8        |
| 2.3 System Users                                     | 8        |
| 2.4 Operating Environment                            | 9        |
| 3. Architecture Diagram                              | 9        |
| 3.1 High-Level Architecture                          | 10       |
| 3.2 Component Responsibilities                       | 10       |
| 3.3 Data Flow or Workflow View                       | 11       |
| 3.4 Module Architecture                              | 12       |
| 3.5 Deployment View                                  | 13       |
| 4. Technology Stack                                  | 14       |
| 4.1 Selected Technologies                            | 14       |
| 4.2 Technology Decisions and Assumptions             | 15       |
| 4.3 Technology Risks                                 | 15       |
| 5. Functional Specifications                         | 17       |
| 5.1 Electronic Visit Verification (EVV) Enhancements | 17       |
| 5.2 AI-Assisted Invoicing (AWS Bedrock + Claude)     | 17       |
| 5.3 QR/Invite Care Circle Onboarding                 | 18       |
| 5.4 Wearable Device Integration                      | 19       |
| 5.5 Home Care Document Digitization                  | 20       |
| 5.6 Screening and Comprehensive Assessment Workflows | 21       |
| 5.7 Cross-Team Functional Dependencies               | 21       |
| 6. API Design Specification                          | 22       |
| 6.1 API Conventions and Standards                    | 22       |
| 6.2 Endpoint Summary                                 | 22       |
| 6.3 Request and Response Examples                    | 23       |
| 6.4 Error Handling and Status Codes                  | 24       |
| 6.5 Administration, Audit, and Reporting APIs        | 24       |
| 6.6 API Versioning and Backward Compatibility        | 24       |
| 7. Data Model Specification                          | 25       |
| 7.1 Data Modeling Approach and Assumptions           | 25       |
| 7.2 Communication Data Models                        | 25       |

|                                                         |    |
|---------------------------------------------------------|----|
| 7.3 Electronic Visit Verification (EVV) Entities        | 26 |
| 7.4 In-Home Residential Support Entities                | 27 |
| 7.5 AI-Assisted Services Entities                       | 28 |
| 7.6 Administration & Audit Entities                     | 30 |
| 7.7 Data Dictionary Alignment & ER Structure            | 31 |
| 8. UI / UX Design Specification                         | 33 |
| 8.1 UI Design Approach and Principles                   | 33 |
| 8.2 Screens and User Workflows Affected                 | 33 |
| 8.3 Wireframes, Mockups, or Screen Notes                | 34 |
| 8.3.1 EVV Dashboard                                     | 34 |
| 8.3.2 AI Invoicing                                      | 34 |
| 8.3.3 QR Onboarding                                     | 35 |
| 8.3.4 Wearables Integration                             | 35 |
| 8.3.5 Document Digitization                             | 36 |
| 8.3.6 Screening and Assessment                          | 36 |
| 8.4 Accessibility Requirements                          | 37 |
| 8.5 Display Text and Localization Impact                | 37 |
| 9. Security Considerations                              | 38 |
| 9.1 Security Controls                                   | 38 |
| 9.1 Data Privacy                                        | 39 |
| 9.2 Authentication and Authorization                    | 39 |
| 9.3 Secure Data Transmission                            | 39 |
| 9.4 API Security                                        | 40 |
| 9.5 Data Storage and Backup                             | 40 |
| 9.6 Logging and Monitoring                              | 40 |
| 9.7 External Service or AI Tool Security                | 40 |
| 9.8 Security Testing                                    | 40 |
| 10. Error Handling                                      | 41 |
| 11. Performance & Scalability                           | 42 |
| 11.1 Objectives                                         | 42 |
| 11.2 Scalability Strategy                               | 42 |
| 11.3 Performance Monitoring                             | 42 |
| 11.4 Bottleneck Mitigation                              | 42 |
| 12. Testing Strategy                                    | 44 |
| 12.1 Testing Overview                                   | 44 |
| 12.2 Testing Levels                                     | 44 |
| 12.2.1 Unit Testing                                     | 44 |
| 12.2.2 Integration Testing                              | 44 |
| 12.2.3 System Testing                                   | 44 |
| 12.2.4 Security Testing                                 | 45 |
| 12.2.5 Accessibility Testing                            | 45 |
| 12.3 Feature-Specific Test Cases                        | 45 |
| 12.3.1 EVV Test Coverage                                | 45 |
| 12.3.2 AI-Assisted Invoice Test Coverage                | 46 |
| 12.3.3 QR Code and Invite-Link Onboarding Test Coverage | 46 |

|                                                      |    |
|------------------------------------------------------|----|
| 12.3.4 Wearable Device Integration Test Coverage     | 46 |
| 12.3.5 Home Care Document Digitization Test Coverage | 46 |
| 12.3.6 Screening and Assessment Test Coverage        | 47 |
| 12.4 Test Automation                                 | 47 |
| 13. Deployment Strategy                              | 48 |
| 13.1 Deployment Environments                         | 48 |
| 13.2 Infrastructure or Hosting                       | 48 |
| 13.3 CI/CD Pipeline                                  | 49 |
| 13.4 Rollback Procedure                              | 49 |
| 14. Known Issues & Limitations                       | 50 |
| 14.1 Technical Limitations                           | 50 |
| 14.2 Known Issues                                    | 50 |
| 14.3 Exclusions                                      | 51 |
| 14.4 Future Enhancements                             | 51 |

# 1. Introduction

## 1.1 Purpose

This Technical Design Document (TDD) defines the technical architecture, implementation approach, integration strategy, and testing considerations for Team D's assigned CareConnect enhancements. The document translates the approved requirements from the Software Requirements Specification (SRS) and Project Plan into a detailed technical design that can guide development, testing, and future maintenance activities.

The purpose of this document is to provide developers, testers, stakeholders, and future maintainers with a clear understanding of how Team D's assigned features will be implemented and integrated within the CareConnect platform. These features include Electronic Visit Verification (EVV) enhancements, AI-assisted invoicing, QR code and invite-link onboarding, wearable device integration, home care document digitization, and assessment workflows.

## 1.2 Intended Audience

This document is intended for multiple stakeholder groups involved in the development, testing, review, and maintenance of the CareConnect platform.

The primary audience includes:

- Software developers responsible for implementing Team D functionality.
- Quality assurance personnel responsible for creating and executing test cases.
- Team leads and project managers responsible for milestone planning and coordination.
- Professor Assadullah and academic reviewers evaluating project progress and technical quality.
- The CareConnect client and stakeholders reviewing proposed solutions and deliverables.
- Future developers who may maintain or extend the platform after the completion of the capstone project.

## 1.3 Document Scope

This document defines the technical design for Team D's assigned feature areas within the CareConnect healthcare coordination platform.

### Included in Scope

- Electronic Visit Verification (EVV) workflow enhancements and integration readiness.
- AI-assisted invoicing workflow migration from DeepSeek to AWS Bedrock utilizing Anthropic Claude.
- QR code and invite-link onboarding workflows for care-circle membership management.
- Wearable device integration supporting Fitbit, Apple HealthKit, and Google Health Connect.
- Home care document digitization workflows.
- Screening and assessment workflow design.
- Supporting APIs, data models, testing considerations, and security requirements associated with these features.

### Excluded from Scope

- Live EVV aggregator submission implementation.
- Production healthcare billing integrations.
- Electronic Health Record (EHR) and FHIR integrations.
- Custom hardware or medical device development.
- AI-based diagnosis or treatment recommendations.
- Enterprise infrastructure deployment and AWS environment configuration.
- Organization-wide RBAC implementation owned by other project teams.

## 1.4 Definitions, Acronyms, and Abbreviations

| Term      | Definition                                                   |
|-----------|--------------------------------------------------------------|
| TDD       | Technical Design Document                                    |
| SRS       | Software Requirements Specification                          |
| STP       | Software Test Plan                                           |
| API       | Application Programming Interface                            |
| UI        | User Interface                                               |
| UX        | User Experience                                              |
| AWS<br>AI | Amazon Web Services<br>Artificial Intelligence               |
| Bedrock   | AWS managed service provide access to foundation AI models   |
| Claude    | Anthropic large language model available through AWS Bedrock |
| EVV       | Electronic Visit Verification                                |
| Fitbit    | Wearable health tracking platform                            |
| HealthKit | Apple health data integration platform                       |
| OCR       | Optical Character Recognition                                |
| PHI       | Protected Health Information                                 |
| QR code   | Quick Response Code user for onboarding workflows            |
| RBAC      | Role-based Access Control                                    |
| REST      | Representational State Transfer                              |

## 1.5 References

| Reference                           | Purpose                                                                       |
|-------------------------------------|-------------------------------------------------------------------------------|
| Software Requirements Specification | Defines approved functional and nonfunctional requirements.                   |
| Project Plan                        | Defines project scope, milestones, responsibilities, risks, and schedule.     |
| Software Test Plan                  | Defines testing strategy, test cases, and acceptance approach.                |
| Cross-Team Collaboration Plan       | Defines team coordination, shared decisions, and conflict resolution process. |

- CareConnect Software Requirements Specification (SRS)
- CareConnect Project Plan
- CareConnect Milestone 3 Coding Readiness Report
- SWEN 670 Course Materials
- IEEE 1016 Software Design Description Guidelines
- AWS Bedrock Documentation
- Fitbit Developer Documentation
- Apple HealthKit Documentation
- Google Health Connect Documentation

# 2. System Overview

## 2.1 System Description

CareConnect is a healthcare coordination platform designed to improve communication, documentation, and workflow management between patients, caregivers, care-circle members, and healthcare administrators. The platform supports healthcare service delivery through secure workflows that enable visit verification, onboarding, documentation management, communication, and operational oversight.

Team D's assigned work focuses on extending and enhancing existing CareConnect capabilities through improvements to Electronic Visit Verification (EVV), AI-assisted invoicing, wearable device integration, QR code and invite-based onboarding workflows, home care document digitization, and assessment workflows. Several of these features already exist in partial form and require enhancement, refinement, or migration rather than complete redevelopment.

The proposed design emphasizes reuse of existing CareConnect architecture, extension of established repository patterns, and minimization of risk associated with external integrations. Where external dependencies such as EVV aggregators, wearable APIs, and AI providers remain unresolved, the system will utilize mock implementations and integration-ready designs until production access and ownership decisions are finalized.

## 2.2 System Objectives

The primary objectives of Team D's technical design are:

1. Enhance existing EVV workflows while maintaining integration readiness for future aggregator connectivity.
2. Support migration of invoice extraction workflows from DeepSeek to AWS Bedrock and Anthropic Claude.
3. Provide secure QR code and invite-link onboarding workflows for care-circle membership management.
4. Support wearable device integration through standardized synchronization and monitoring workflows.
5. Digitize home care documentation to reduce reliance on paper-based processes.
6. Establish structured workflows for screening and comprehensive assessments.
7. Protect PHI through secure design practices and role-based access controls.
8. Reuse existing CareConnect architecture and repository patterns whenever feasible.
9. Minimize technical risk by isolating unresolved external dependencies behind integration boundaries.
10. Support future expansion through modular, maintainable, and scalable system design.
11. Enable efficient testing through clearly defined interfaces, workflows, and integration points.
12. Ensure alignment with the approved SRS, Project Plan, and milestone deliverables.

## 2.3 System Users

| User Type | Relevant Needs                                        | Design Impact                                                                               |
|-----------|-------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Caregiver | Performs health visit; submits EVV check-in/check-out | EVV gap closure; wearable sync consent; invite receipt; assessment and screening submission |



|                      |                                                                        |                                                                                            |
|----------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Patient/Client       | Receives home care services; may share wearable biometric data         | Wearable metrics target; screening and assessment subject                                  |
| Supervisor           | Manages caregiver assignments; reviews EVV records and assessments     | EVV aggregator dashboard; invoice review; assessment read access                           |
| Billing Specialist   | Processes invoices; reviews Bedrock-extracted field data               | Invoice extraction workbench; field correction; approval workflow                          |
| Administrator        | Full platform access; manages users, invites, and system configuration | Invite generation and revocation; aggregator configuration; document digitization approval |
| New Hire Coordinator | Onboards new caregivers; collects I-9, W-4, and Direct Deposit forms   | Home care document digitization design; screening intake workflow                          |

## 2.4 Operating Environment

| Environment Area     | Expected Description                                                                                                                                                                                                                                                            |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Web Client           | Flutter Web application running in Chrome, Firefox, and Edge. Responsive layout targeting desktop and tablet viewports                                                                                                                                                          |
| iOS Client           | Flutter iOS application deployed on iPhone and iPad. Uses APNs for push notifications. Apple HealthKit wearable integration is deferred to Milestone 3 pending Apple developer entitlement confirmation                                                                         |
| Android Client       | Flutter Android application; primary mobile target for Team D. Integrated with Google Health Connect SDK (androidx.health.connect.client) for on-device biometric data access. No biometric data transmitted to Google servers.                                                 |
| Backend Server       | Java 17 with Spring Boot 3.4.5 serving a RESTful API. Stateless service architecture. Spring Security with JWT/OAuth2 handles authentication and RBAC enforcement using role constants provided by Team A.                                                                      |
| Database             | PostgreSQL 15.x hosted on Amazon RDS. Accessed via Spring Data JPA. Field-level AES-256 encryption at rest managed through AWS KMS.                                                                                                                                             |
| AI/ML Inference      | AWS Bedrock (Anthropic Claude) used for invoice field extraction. Pending Team A provisioning of Anthropic account credentials. MockBedrockAdapter substitutes in Milestone 2 and returns predefined structured test data                                                       |
| Document Storage     | Amazon S3 with HIPAA-compliant bucket policies. All invoice attachments and home care documents stored with server-side encryption (SSE-KMS).                                                                                                                                   |
| Wearable Integration | Google Health Connect SDK (androidx.health.connect.client) on Android. On-device API only; no biometric data is transmitted to Google servers. Supports heart rate, steps, and blood oxygen metric types. Android-only in Milestone 2; Apple HealthKit targeted in Milestone 3. |
| Push Notifications   | Firebase Cloud Messaging (FCM) for Android wearable threshold alerts. Apple Push Notifications (APNs) for iOS. Both services managed by Team A infrastructure.                                                                                                                  |
| Secrets Management   | AWS Secrets Manager stores all API keys, database credentials, and the HMAC signing key used for invite token generation and validation.                                                                                                                                        |

## 3. Architecture Diagram

This section documents the backend and integration architecture for Team D's Milestone 2 scope within CareConnect. Team D extends the shared platform with a Spring Boot service foundation, relational persistence, role-protected APIs, AI-assisted invoice extraction, EVV readiness workflows, secure invite and QR onboarding, wearable integration boundaries, structured document capture, and assessment-related backend support. The

design emphasizes safe extension of the existing repository, separation of provider-specific adapters from business logic, and mock-first validation when live external dependencies remain unapproved or unavailable.

### 3.1 High-Level Architecture

At a high level, the Team D implementation follows a layered application model. The Flutter client invokes protected backend APIs over HTTPS. Requests first traverse shared authentication and role enforcement boundaries, then flow into Spring Boot controllers and service classes that apply validation, workflow rules, and persistence logic. External integrations such as AWS Bedrock, OCR/extraction services, EVV aggregator clients, and wearable provider adapters remain behind dedicated service boundaries so the user interface and domain rules are not tightly coupled to any single provider implementation.

Core backend components referenced across the attached materials include EvvController.java, EvvService.java, HhaExchangeBatchSubmissionService.java, InvoiceController.java, TextractService.java, LlmExtractionService.java, LinkManagementController.java, and FitbitService.java. These components fit the existing controller-service-client pattern already visible in the repository and readiness artifacts.

Figure 1. Team D High-Level Architecture

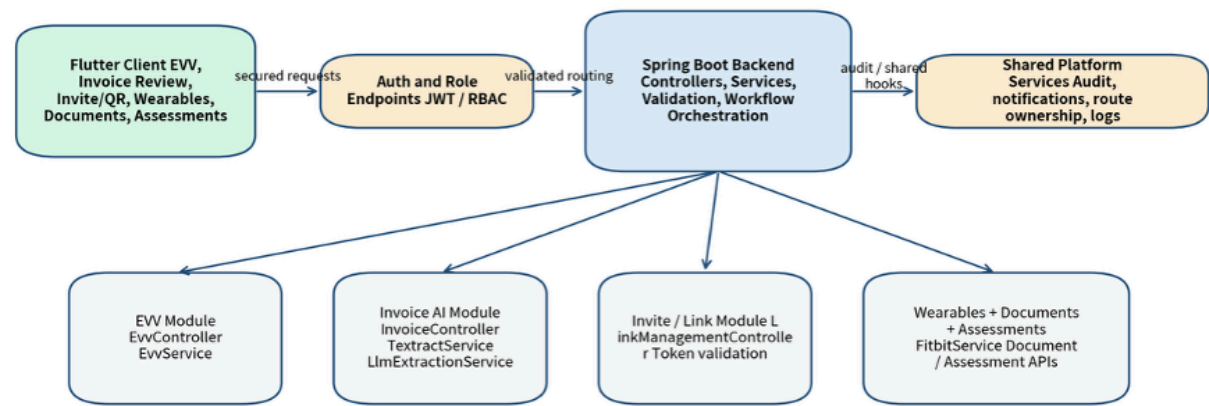


Figure 1. Team D high-level backend architecture.

### 3.2 Component Responsibilities

| Component                | Responsibility                                                                                                                        | Owner                          | Dependencies                                 |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------|
| Flutter Client Workflows | Captures user actions for EVV, invoices, invites, wearables, documents, and assessments; displays backend responses and review states | Frontend / shared client layer | Auth and role endpoints; Team D backend APIs |

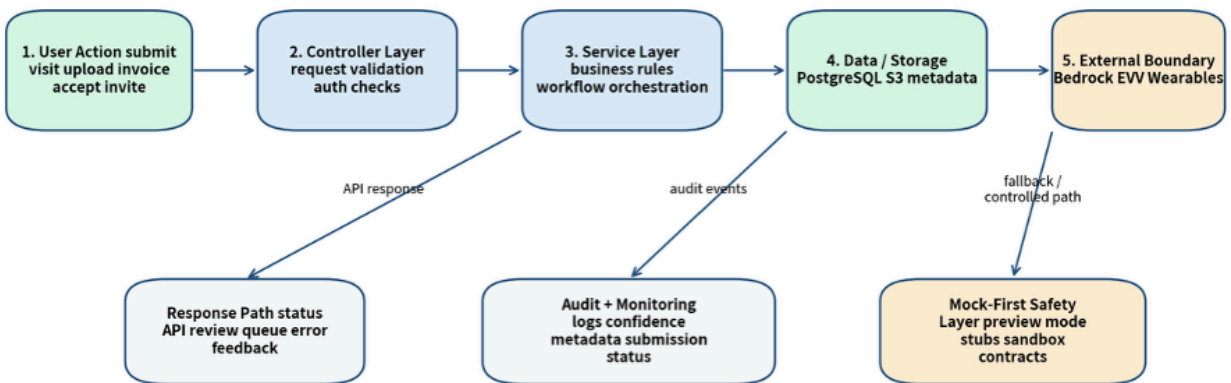
| Component                                | Responsibility                                                                                                      | Owner                                           | Dependencies                                                           |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------|
| Auth and Role Endpoints                  | Apply JWT and RBAC controls before protected workflow access is granted                                             | Shared platform ownership with Team A alignment | Role matrix, route ownership, token handling                           |
| InvoiceController.java                   | Receives invoice upload and extraction requests; coordinates extraction-review workflow entry points                | Team D backend                                  | TextextractService, LlmExtractionService, storage, invoice status API  |
| TextextractService.java                  | Performs OCR-oriented extraction or document preprocessing before LLM mapping                                       | Team D backend                                  | Uploaded file handling, provider boundary, extraction contract         |
| LlmExtractionService.java                | Maps extracted text to structured invoice fields using LLM-assisted reasoning and confidence handling               | Team D backend                                  | AWS Bedrock / Claude, review workflow, audit metadata                  |
| EvvController.java / EvvService.java     | Manage EVV lifecycle operations, validation, correction, and readiness visibility                                   | Team D backend                                  | Visit records, aggregator status API, submission preparation           |
| HhaExchangeBatchSubmissionService.java   | Prepares outbound EVV payloads and isolates submission logic behind a controllable boundary                         | Team D backend                                  | VirginiaHhaExchangeClient, DcSandataAltEvvClient, mock or sandbox mode |
| LinkManagementController.java            | Supports invite token issuance, acceptance, QR/link validation, and related onboarding flows                        | Team D backend                                  | Auth and role endpoints, caregiver/patient link APIs                   |
| FitbitService.java and wearable adapters | Normalize provider payloads, support sync boundaries, and isolate external wearable integrations                    | Team D backend                                  | OAuth adapters, metric ingestion service, mock payloads                |
| Persistence Layer                        | Stores visit records, invoice drafts, invite tokens, document states, assessment states, and wearable sync metadata | Shared database / Team D feature usage          | PostgreSQL, Spring Data JPA, approved schema paths                     |
| Shared Platform Services                 | Provide logging, monitoring, audit hooks, storage, and deployment infrastructure                                    | Shared platform ownership                       | CloudWatch, S3, IAM, KMS, CI/CD pipeline                               |

### 3.3 Data Flow or Workflow View

The primary Team D workflows follow a common backend pattern: a user action triggers a protected API call, the controller validates input and user access, the service layer applies business rules, data is persisted or updated, and any external provider interaction is routed through a controlled adapter boundary. The same structural pattern is reused for EVV lifecycle changes, invoice extraction and review, invite acceptance, wearable synchronization, and document or assessment status changes.

| Step | Action                                                                                                        | System Response                                                                                           |
|------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1    | User initiates a protected workflow such as EVV update, invoice upload, invite acceptance, or assessment save | Flutter UI invokes the appropriate Team D API over HTTPS                                                  |
| 2    | Authentication and role validation occur                                                                      | Only authorized users proceed to controller handling; invalid or unauthorized requests are rejected early |
| 3    | Controller layer validates request structure and forwards work to the appropriate service                     | Malformed input is rejected and safe error feedback is returned                                           |
| 4    | Service layer applies workflow rules and persistence operations                                               | Domain state is updated in PostgreSQL and any file metadata or audit details are recorded                 |
| 5    | Optional external-service processing occurs behind a provider boundary                                        | Bedrock, OCR, EVV, or wearable interactions use mock, sandbox, or approved non-production paths           |
| 6    | Status, review state, or preview output is returned to the client                                             | User sees success state, review queue, corrected data, or controlled failure message                      |

**Figure 2. Backend Data Flow and Workflow View**



*Team D backend data flow and workflow view.*

### 3.4 Module Architecture

Internally, the Team D backend follows a modular structure that separates controllers, domain services, persistence logic, and external adapters. Controllers define route-level API entry points and enforce request-shape expectations. Services contain business rules, status transitions, and orchestration across

repositories or integration clients. Repository and persistence concerns remain focused on structured storage of visits, draft records, tokens, metadata, and workflow states. Provider-specific integration logic is isolated in adapters or client classes so contract changes do not force redesign of the domain layer.

This architecture supports six major feature families within Team D scope: EVV enhancements, invoice AI extraction and review, invite/QR onboarding, wearable integration boundaries, document digitization support, and assessment workflow support. Even where production integrations are not yet approved, the module boundaries remain stable because mock responses and preview modes use the same internal service contracts as future production-safe implementations.

### 3.5 Deployment View

Team D backend features run across three environments: local development, shared test validation, and final demonstration. Local development uses Docker-compatible or local Spring Boot execution with seeded data and mock integrations. The test environment supports pre-integration validation and approved non-production service access when available. The final demonstration environment uses shared hosted infrastructure and only approved provider contracts. This approach allows Team D to validate application behavior early while reducing the risk of rework caused by unstable or unapproved external dependencies.

Figure 3. Deployment View

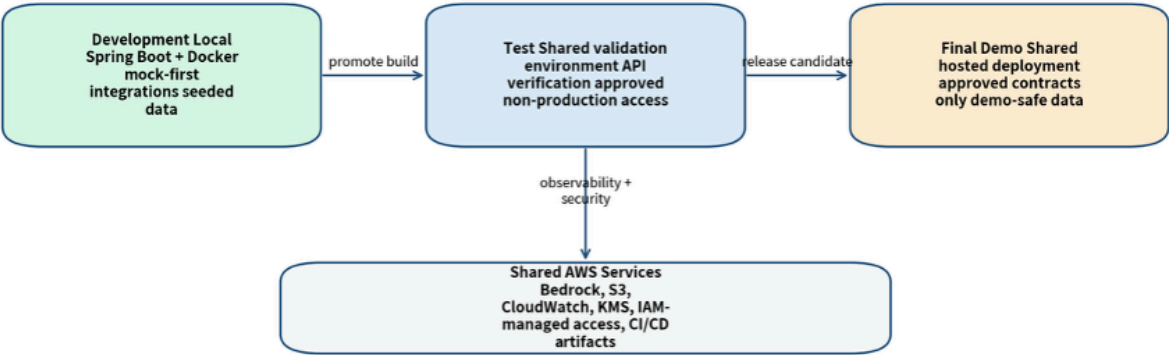


Figure 3. Deployment view for Team D backend components.

## 4. Technology Stack

The Team D backend stack was selected to align with the shared CareConnect architecture, support maintainable service-layer design, and enable secure integration with managed cloud services and external provider boundaries.

### 4.1 Selected Technologies

| Layer                | Technology                                            | Purpose                                                                                          | Status / Owner                                    |
|----------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Backend              | Java 17 + Spring Boot 3.4.5                           | Primary application framework for controllers, services, validation, and modular backend design  | In use / Team D backend                           |
| Persistence          | PostgreSQL + Spring Data JPA                          | Relational storage for visits, invoice drafts, tokens, documents, assessments, and sync metadata | In use / Shared database + Team D feature use     |
| AI Inference         | AWS Bedrock + Claude                                  | LLM-assisted invoice field extraction and structured response mapping                            | Approved dependency path with Team A coordination |
| OCR / Extraction     | Textract-aligned extraction service pattern           | Text capture and preprocessing before LLM-assisted field mapping                                 | In use / Team D backend service boundary          |
| API Contract         | REST-style HTTPS APIs                                 | Expose Team D backend workflows to the Flutter client and shared platform consumers              | In use / Shared application pattern               |
| Identity / Access    | JWT / OAuth2 / RBAC integration                       | Protect restricted backend actions and enforce role-appropriate workflow access                  | Shared platform ownership with Team D integration |
| Object Storage       | Amazon S3 or equivalent object-storage pattern        | Store uploaded invoice and document artifacts outside relational tables                          | Shared platform service                           |
| Monitoring / Logging | CloudWatch-compatible centralized logging and metrics | Operational visibility, troubleshooting, and deployment validation                               | Shared platform service                           |
| Encryption / Secrets | KMS and IAM-managed access                            | Protect credentials, service access, and sensitive operational configuration                     | Shared platform service                           |
| Client Interface     | Flutter + Dart                                        | Consumes backend APIs and presents EVV, invoice, onboarding, and related workflow states         | Shared frontend / Team D coordination             |

## 4.2 Technology Decisions and Assumptions

| Decision or Assumption                                      | Rationale                                                                                                         | Impact                                                                                         |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Reuse the shared Spring Boot architecture                   | The existing repository already contains controller, service, and integration patterns relevant to Team D scope   | Reduces duplication and keeps backend implementation consistent with the rest of CareConnect   |
| Keep provider logic behind adapters or clients              | External providers such as Bedrock, EVV aggregators, and wearable APIs may change independently of domain logic   | Allows contract updates without redesigning the core service layer                             |
| Use human-reviewed AI output only                           | Invoice extraction is informational support and must be validated by a user before final acceptance               | Improves safety, auditability, and trust in extracted data                                     |
| Adopt mock-first integration for unresolved dependencies    | Several live integrations depend on credentials, route ownership, governance decisions, or sandbox availability   | Allows development and testing progress without enabling unsupported production calls          |
| Use relational persistence for workflow state and metadata  | Visits, draft records, tokens, and approval states require consistent transactions and queryable relationships    | Supports traceability and repeatable state management                                          |
| Store large files outside relational core tables            | Uploaded invoices and similar artifacts are more efficiently managed in object storage                            | Reduces database bloat and simplifies file lifecycle handling                                  |
| Rely on Team A for base CI/CD and shared RBAC ownership     | The project plan and pre-code summary assign core deployment pipeline and RBAC framework ownership outside Team D | Team D focuses on feature-complete artifacts that integrate cleanly into shared infrastructure |
| Avoid new assessment or home-care migrations until approved | Readiness artifacts identify Flyway and migration-numbering constraints                                           | Prevents schema conflicts and keeps milestone work inside approved database scope              |

## 4.3 Technology Risks

| Risk                                      | Impact                                                                                                | Mitigation                                                                                           |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| EVV aggregator target or credential delay | Live EVV submission may remain unavailable, limiting validation to payload preview or mock submission | Maintain a preview-only or mock-first EVV boundary and validate internal lifecycle correctness first |
| Bedrock / IAM provisioning delay          | Provider-backed invoice extraction could be blocked even if internal review workflow is complete      | Use controlled mock extraction responses that preserve the final contract shape                      |

| Risk                                           | Impact                                                                                                   | Mitigation                                                                             |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Wearable provider sandbox or rate-limit issues | Live wearable sync validation may be incomplete or unstable                                              | Use adapter isolation, sample payload replay, and graceful degradation behavior        |
| Migration and Flyway conflicts                 | Unapproved schema work could break shared environments or create merge conflicts                         | Constrain implementation to approved schema paths and document deferred migration work |
| RBAC or route ownership changes                | Authorization behavior could drift across teams and invalidate endpoint assumptions                      | Coordinate with shared route ownership and role matrices before final integration      |
| Improper AI or sensitive-data handling         | Uploading protected data or trusting AI outputs without review could create security and compliance risk | Use sanitized test data, human review, and protected-route enforcement                 |



## 5. Functional Specifications

### 5.1 Electronic Visit Verification (EVV) Enhancements

| Item                     | Team Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requirement IDs or Names | REQ-5.1.1-001, REQ-5.1.1-002, REQ-5.1.1-003, REQ-5.1.2-001, REQ-5.1.2-002, REQ-5.1.2-003, REQ-5.1.3-001, REQ-5.1.3-002, REQ-5.1.3-003,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Feature Summary          | Team D enhances EVV workflow reliability across three areas: visit lifecycle gap closure and correction (5.1.1), aggregator dependency tracking and operational visibility (5.1.2), and EVV submission design with error handling readiness (5.1.3). Together these ensure that EVV records are complete, aggregator-ready, and auditable from scheduling through downstream submission handoff.                                                                                                                                                                                                                                                                                                                                                    |
| Primary User Flow        | <ol style="list-style-type: none"> <li>1. Caregiver or admin opens the EVV dashboard and selects a visit record.</li> <li>2. User advances the visit through lifecycle states: scheduled, check-in, in-progress, check-out, completed.</li> <li>3. System validates required fields at each transition and surfaces missing or invalid data.</li> <li>4. For aggregator dependency: admin views aggregator readiness status and blocking conditions in operational view before submission.</li> <li>5. User corrects identified field issues and retries the failed transition or submission preparation step.</li> <li>6. System validates and confirms readiness, then queues the record for downstream EVV aggregator or EDI handoff.</li> </ol> |
| Main System Behaviors    | Supports all EVV visit lifecycle state transitions from scheduling through completion. Enforces required field validation before state progression is allowed. Supports non-destructive correction and retry of failed transitions and submission preparation steps. Surfaces aggregator dependency status and blocking conditions in administrative views. Validates required EVV data elements before export or submission readiness is confirmed. Preserves timestamped, auditable records of all state changes, corrections, and submission preparation outcomes.                                                                                                                                                                               |
| Inputs                   | Visit identifiers, caregiver and user workflow actions, EVV field updates, aggregator status signals, submission preparation triggers, export configuration values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Outputs                  | [Updated EVV visit record states, field validation feedback, aggregator readiness indicators, retry outcomes, submission preparation results, EDI/handoff queue entries, and audit event records.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Alternate / Error Flows  | If a required field is missing at state transition: system blocks progression and surfaces a field-level error with recovery guidance. If aggregator is unavailable or blocking: admin view reflects the condition and prevents submission until resolved. If submission preparation fails: system routes the record to a correctable error state, preserves existing data, and provides actionable failure context for retry.                                                                                                                                                                                                                                                                                                                      |

### 5.2 AI-Assisted Invoicing (AWS Bedrock + Claude)

| Item                     | Team Response                                                                                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requirement IDs or Names | REQ-5.2.1-001, REQ-5.2.1-002, REQ-5.2.1-003, REQ-5.2.2-001, REQ-5.2.2-002, REQ-5.2.2-003, REQ-5.2.3-001, REQ-5.2.3-002, REQ-5.2.3-003                                     |
| Feature Summary          | Team D migrates the invoice AI extraction provider from the legacy path to AWS Bedrock Claude while preserving existing intake behavior and reviewer workflow continuity. |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | The feature covers invoice intake and extraction (5.2.1), reviewer validation and approval (5.2.2), and Bedrock model configuration and controlled failure handling (5.2.3).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Primary User Flow       | <ol style="list-style-type: none"> <li>1. Billing user uploads an invoice image or document through the intake screen.</li> <li>2. System submits the invoice content through the Bedrock Claude extraction path and returns structured field output.</li> <li>3. Reviewer opens the extracted invoice draft and inspects extracted fields alongside confidence context.</li> <li>4. Reviewer corrects any inaccurate or missing fields.</li> <li>5. Reviewer approves the invoice or routes it for manual handling if extraction quality is insufficient.</li> <li>6. System stores the final reviewed invoice with extraction metadata and approval record.</li> </ol> |
| Main System Behaviors   | Processes invoice extraction through AWS Bedrock Claude for all Team D invoice workflows. Returns structured invoice fields required for reviewer validation including confidence metadata. Preserves the existing upload-review-approve workflow continuity after provider migration. Presents extracted fields as editable before final approval is permitted. Validates Bedrock extraction configuration before processing invoice requests. Routes failed or low-confidence extractions to a manual review queue with failure context preserved. Maintains traceability of AI-assisted extractions versus manual reviewer corrections.                               |
| Inputs                  | Invoice image or document file, Bedrock model configuration, reviewer field edits, approval or manual-route actions, model and schema configuration state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Outputs                 | Structured invoice draft with extracted fields and confidence indicators, reviewer correction records, approval status, failure or retry messages, manual queue entries, and audit log entries.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Alternate / Error Flows | If extraction confidence is below threshold: system creates draft with low-confidence warnings and routes to mandatory manual review. If Bedrock configuration is invalid or unavailable: system blocks extraction, records the failure context, and routes the invoice to the manual queue. If a transient provider error occurs: system marks the request as retrievable with original content preserved for resubmission.                                                                                                                                                                                                                                             |

### 5.3 QR/Invite Care Circle Onboarding

| Item                     | Team Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requirement IDs or Names | REQ-5.3.1-001, REQ-5.3.1-002, REQ-5.3.1-003, REQ-5.3.2-001, REQ-5.3.2-002, REQ-5.3.2-003, REQ-5.3.3-001, REQ-5.3.3-002, REQ-5.3.3-003                                                                                                                                                                                                                                                                                                                                                                                        |
| Feature Summary          | Team D delivers secure invite-based onboarding that allows existing users to bring new participants into a care circle through tokenized URL links and QR codes. The feature covers invite token issuance and delivery (5.3.1), invite acceptance and role assignment (5.3.2), and token lifecycle and exception handling (5.3.3).                                                                                                                                                                                           |
| Primary User Flow        | <ol style="list-style-type: none"> <li>1. Inviter initiates invite creation and selects the intended recipient role.</li> <li>2. System generates a unique, time-limited invite token and delivers it as a URL and QR code.</li> <li>3. Recipient opens the invite link or scans the QR code to begin onboarding.</li> <li>4. System validates the token state (valid, expired, revoked, already used) before allowing progression.</li> <li>5. Recipient completes registration and identity verification steps.</li> </ol> |

| Item                    | Team Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | 6. System assigns the appropriate care circle membership and role scope and routes the new user to their role dashboard.                                                                                                                                                                                                                                                                                                                                                                                          |
| Main System Behaviors   | Generates unique, time-limited invitation tokens for care circle onboarding. Delivers tokens as both URL links and QR codes. Validates token state before allowing any onboarding progression. Assigns care circle membership and role scope upon successful acceptance. Blocks onboarding for invalid, expired, revoked, or already-used tokens. Prevents duplicate onboarding from reused invitation tokens. Preserves onboarding session continuity and provides guided recovery messaging for token failures. |
| Inputs                  | Invite creation request, recipient role selection, token payload from URL or QR scan, recipient registration and verification details, token validation outcomes.                                                                                                                                                                                                                                                                                                                                                 |
| Outputs                 | Invite URL and QR code artifacts, token validation result, onboarding status, role-linked care circle account association, token lifecycle events, and session routing to role dashboard.                                                                                                                                                                                                                                                                                                                         |
| Alternate / Error Flows | If token is expired: system blocks progression and presents expiration messaging with guidance to request a new invite. If token is already used: system blocks progression and prevents duplicate account creation. If token is revoked: system displays a revocation message and provides contact guidance. If registration fails mid-flow: system preserves the session context so the user can retry without requiring a new token.                                                                           |

## 5.4 Wearable Device Integration

| Item                     | Team Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requirement IDs or Names | REQ-5.4.1-001, REQ-5.4.1-002, REQ-5.4.1-003, REQ-5.4.2-001, REQ-5.4.2-002, REQ-5.4.2-003, REQ-5.4.3-001, REQ-5.4.3-002, REQ-5.4.3-003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Feature Summary          | Team D delivers wearable device integration covering secure connector authorization and account linking (5.4.1), metric synchronization, normalization, and dashboard visibility (5.4.2), and threshold alert management and sync failure notifications (5.4.3).                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Primary User Flow        | <ol style="list-style-type: none"> <li>1. User navigates to wearable integration settings and selects a supported connector to link.</li> <li>2. System initiates the provider authorization flow and completes secure account linking.</li> <li>3. Wearable metrics begin syncing and are normalized into the dashboard view with latest-sync timestamps.</li> <li>4. Admin or care coordinator configures threshold alert rules for supported metric types.</li> <li>5. System evaluates incoming metrics against configured thresholds and sync freshness limits.</li> <li>6. User or admin receives alert notifications and takes acknowledgment or escalation actions.</li> </ol> |
| Main System Behaviors    | Supports secure authorization workflows for configured wearable connectors. Persists account link state and connection health status. Ingests and normalizes supported wearable metric payloads. Displays latest sync timestamps and connection health in dashboard views. Supports configurable threshold alert rules for supported wearable metrics. Triggers alert and notification workflows when thresholds are breached or sync freshness falls below configured limits. Supports alert acknowledgment and escalation actions.                                                                                                                                                   |

| Item                    | Team Response                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inputs                  | Connector authorization responses, wearable metric payloads, threshold configuration rules, sync freshness limits, alert acknowledgment actions.                                                                                                                                                                                                                                                             |
| Outputs                 | Account link status, normalized metric display, sync timestamps, threshold alert events, notification deliveries, acknowledgment records, and escalation state.                                                                                                                                                                                                                                              |
| Alternate / Error Flows | If connector authorization fails: system displays failure context and provides retry or re-authorization guidance without losing existing link state. If metric sync is stale beyond the configured limit: system surfaces a sync freshness warning and prompts reconnect. If a threshold alert is not acknowledged within a defined window: system escalates the alert to the appropriate care team member. |

## 5.5 Home Care Document Digitization

| Item                     | Team Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requirement IDs or Names | REQ-5.5.1-001, REQ-5.5.1-002, REQ-5.5.1-003, REQ-5.5.2-001, REQ-5.5.2-002, REQ-5.5.2-003, REQ-5.5.3-001, REQ-5.5.3-002, REQ-5.5.3-003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Feature Summary          | Team D digitizes home care documentation into structured, validated intake forms with a full document lifecycle and profile-linked traceability. The feature covers structured form digitization and validation (5.5.1), document review states and administrative actions (5.5.2), and traceability and profile-linked persistence (5.5.3).                                                                                                                                                                                                                                                                                            |
| Primary User Flow        | <ol style="list-style-type: none"> <li>1. User opens a digital home care form from the document intake screen.</li> <li>2. User completes required structured fields; system validates required entries in real time.</li> <li>3. User saves the form as a draft for later completion or submits for review.</li> <li>4. Reviewer or admin opens the submitted document and inspects content.</li> <li>5. Admin accepts the document or returns it with correction guidance.</li> <li>6. System records the lifecycle state change with actor identity and timestamp and links the document to the appropriate user profile.</li> </ol> |
| Main System Behaviors    | Provides structured digital intake forms for approved home care document types. Validates required fields before draft save and enforces completion rules before submission. Supports draft, submitted, pending review, and accepted or returned document lifecycle states. Allows authorized reviewers to accept or return submitted documents. Persists digitized document records linked to the appropriate user profile. Preserves status history and actor and timestamp metadata for all document state changes. Maintains source-to-structured traceability for audited document review.                                         |
| Inputs                   | Structured form field data, document type selection, save or submit actions, reviewer accept or return decisions, reviewer comments, profile context.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Outputs                  | Structured document records, lifecycle status, review decisions, correction guidance, traceability metadata, actor and timestamp audit records.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Alternate / Error Flows  | If required fields are missing at submission: system blocks submission and highlights incomplete fields with actionable guidance. If a document is returned by the reviewer: system notifies the submitter with correction context and allows re-submission without creating a duplicate record. If a draft                                                                                                                                                                                                                                                                                                                             |

| Item | Team Response                                                                                  |
|------|------------------------------------------------------------------------------------------------|
|      | is abandoned: system retains the draft record linked to the user profile for later resumption. |

## 5.6 Screening and Comprehensive Assessment Workflows

| Item                     | Team Response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Requirement IDs or Names | REQ-5.6.1-001, REQ-5.6.1-002, REQ-5.6.1-003, REQ-5.6.2-001, REQ-5.6.2-002, REQ-5.6.2-003, REQ-5.6.3-001, REQ-5.6.3-002, REQ-5.6.3-003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Feature Summary          | Team D delivers structured clinical intake workflows covering initial screening capture (5.6.1), comprehensive multi-domain assessment capture (5.6.2), and risk scoring, submission controls, and escalation readiness (5.6.3).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Primary User Flow        | <ol style="list-style-type: none"> <li>1. Assessor opens a new screening or comprehensive assessment from the patient profile.</li> <li>2. System loads the appropriate structured form sections and required field rules for the assessment type.</li> <li>3. Assessor completes sections and saves a draft for later resumption as needed.</li> <li>4. Assessor records disposition or risk context as part of assessment completion.</li> <li>5. Assessor submits the completed assessment; system validates required fields and records submission.</li> <li>6. System generates follow-up action prompts based on risk or disposition context and persists the assessment linked to patient and assessor metadata.</li> </ol> |
| Main System Behaviors    | Supports structured screening fields required for intake and disposition decisions. Supports comprehensive assessment domains including clinical, social, functional, and legal history. Associates each screening and assessment record with patient identity and assessor metadata. Supports save-as-draft and resume for partially completed assessment sessions. Supports final submission with required field enforcement. Preserves risk and disposition metadata associated with screening and assessment outcomes. Generates escalation or follow-up action context based on outcome rules. Persists completed assessments in patient records for longitudinal retrieval.                                                  |
| Inputs                   | Assessment form responses per structured domain, assessor identity and time context, patient profile linkage, disposition and risk values, draft save and final submission actions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Outputs                  | Persisted screening and assessment records, submission status, risk and disposition metadata, follow-up or escalation action context, and longitudinal patient record updates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Alternate / Error Flows  | If required fields are incomplete at submission: system blocks final submission and surfaces field-level guidance per section. If a session is interrupted: system preserves draft state so the assessor can resume without data loss. If elevated risk is captured: system presents escalation prompt and requires explicit disposition acknowledgment before the assessment can be closed.                                                                                                                                                                                                                                                                                                                                       |

## 5.7 Cross-Team Functional Dependencies

| Dependency                                                             | Teams or Owners Involved | Decision Needed                                                      | Status      |
|------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------|-------------|
| AWS Bedrock Claude extraction service availability and IAM role access | Team A                   | Confirm Bedrock endpoint and model ID are provisioned and accessible | In progress |

|  |  |                              |  |
|--|--|------------------------------|--|
|  |  | for Team D invoice workflows |  |
|--|--|------------------------------|--|

## 6. API Design Specification

### 6.1 API Conventions and Standards

- The CareConnect API adheres to strictly governed RESTful principles. All external API traffic is routed over HTTPS utilizing **TLS 1.3** (mandatory per SRS §6.1; Project Plan baseline of TLS 1.2+ is superseded by SRS for compliance). Plaintext HTTP is rejected at both the API gateway and the backend. Payloads strictly utilize JSON. Authentication and authorization are enforced via OAuth 2.0 with JWT bearer tokens (signed with **HS256 or RS256** per SRS) passing through the Spring Security filter chain.
- **Semantic Resource Naming:** Endpoints utilize standard nouns and hierarchy (e.g., `/api/v1/patients/{id}/visits`).
- **HTTP Verbs:** Standard mapping of GET, POST, PUT, PATCH, and DELETE.
- **Idempotency:** All state-mutating requests (POST/PUT/PATCH) accept an `Idempotency-Key` header to prevent duplicate processing.
- **Pagination & Rate Limiting:** Standard cursor-based pagination is implemented for list endpoints. Token bucket rate-limiting ensures fairness and mitigates abuse.
- **Tracing:** `X-Correlation-ID` is required for all requests to ensure distributed tracing via AWS X-Ray .
- Identify privacy or sensitive-data restrictions.
- Write Not applicable if the assigned work does not use or design APIs.

### 6.2 Endpoint Summary

Communication APIs handle secure messaging, telehealth sessions, and alerts. Real-time chat is facilitated through WebSocket channels, while audio/video sessions are bootstrapped via the AWS Chime SDK. Push notifications are routed through Firebase Cloud Messaging (FCM).

| Endpoint                                | Method | Description                                 | Security Context       |
|-----------------------------------------|--------|---------------------------------------------|------------------------|
| <code>/api/v1/messages/send</code>      | POST   | Send a secure text message to a care circle | JWT + Care Circle RBAC |
| <code>/api/v1/telehealth/session</code> | POST   | Bootstrap an AWS Chime audio/video session  | JWT + Telehealth Auth  |

|             |     |                                          |                       |
|-------------|-----|------------------------------------------|-----------------------|
| /ws/v1/chat | WSS | WebSocket channel for real-time delivery | JWT Token Negotiation |
|-------------|-----|------------------------------------------|-----------------------|

### 6.3 Request and Response Examples

Electronic Visit Verification (EVV) APIs comply with the 21st Century Cures Act §12006. Team D's scope focuses on gap closure and enhancement of the existing EVV lifecycle (per SRS §3.14), not greenfield rebuild of check-in/out flows.

- **Visit Lifecycle:** Endpoints expose end-to-end EVV state transitions from *scheduled* → *in-progress* → *completed* → *submitted*.
- **Correction & Retry:** Caregivers and supervisors can amend incomplete or invalid EVV records, with full audit retention.
- **Current-State Review API:** Internal endpoints expose existing field/workflow inventories used by the gap analysis.
- **Feature Enhancements:** New endpoints implement gaps identified in the review (e.g., refined GPS validation, late-submission handling).
- **Pre-Export Validation:** All required EVV data elements are validated before submission readiness.
- **EDI Aggregator Export:** Generates EDI-packaged export batches per documented field mapping for external aggregator ingestion. *Note: SRS specifies generic “EDI exports”; the specific transaction set (e.g., 837P) is to be confirmed with the target aggregator.*

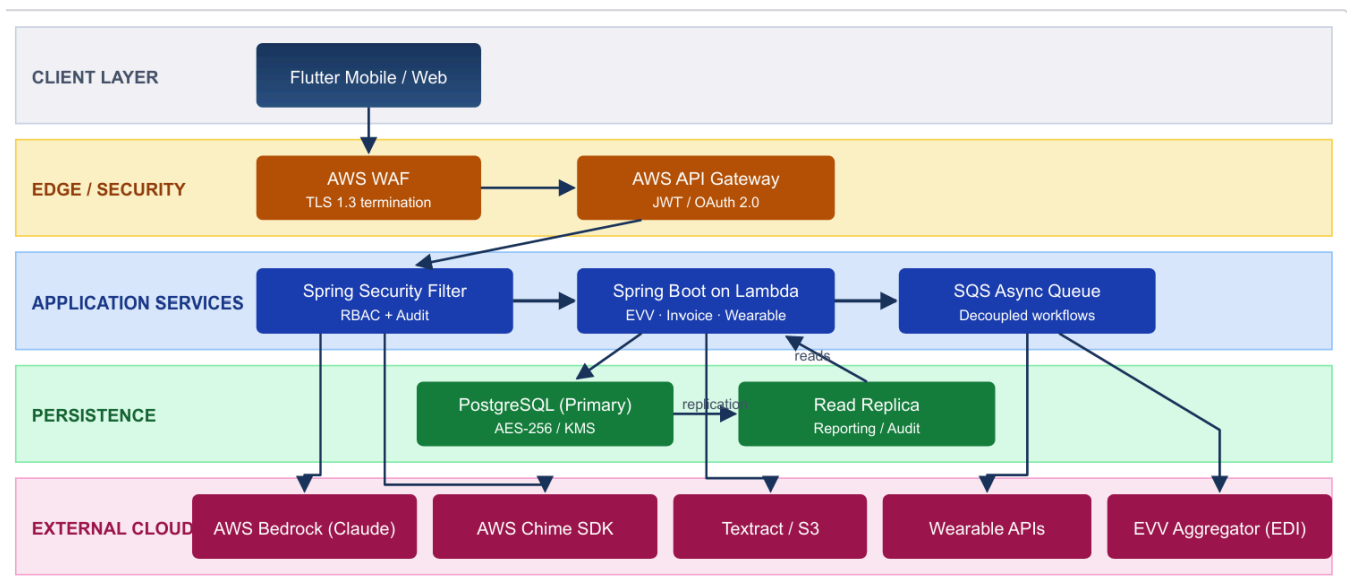


Figure 1. API request flow — client traffic terminates TLS 1.3 at the AWS WAF edge, is admitted by AWS API Gateway, authenticated by the Spring Security filter, and routed to stateless Spring Boot services on AWS Lambda. Writes target the PostgreSQL primary (with asynchronous replication to a read replica used for reporting/audit reads); asynchronous workflows decouple via AWS SQS. External cloud calls (Bedrock, Chime,

Textract/S3, wearable APIs, EVV aggregator) are routed around the persistence layer to keep traffic paths visually distinct.



## 6.4 Error Handling and Status Codes

| Status / Error | Meaning                 | System Response                                 |
|----------------|-------------------------|-------------------------------------------------|
| 400            | Validation error        | Return clear correction guidance.               |
| 401            | Unauthenticated         | Prompt the user to log in.                      |
| 403            | Unauthorized            | Block the action and explain access limitation. |
| 404            | Resource not found      | Show not found message or safe fallback.        |
| 500            | Server or service error | Show safe generic message and log details.      |
| [Other]        | [Meaning]               | [Response]                                      |

## 6.5 Administration, Audit, and Reporting APIs

Supports operational governance. Includes user/role provisioning, EVV state submission tracking, and immutable audit log querying. Reports are strictly generated in a secondary read-replica environment to prevent primary DB contention.

## 6.6 API Versioning and Backward Compatibility

All APIs utilize URI versioning (e.g., /api/v1/). Deprecated endpoints will return a Deprecation header. A strict 12-month sunset policy applies to all endpoints. Pact is the adopted contract-testing tool (Team D extended decision), ensuring client/server compatibility during CI/CD pipelines. Per Project Plan v2.0 §3.2.9, Team A owns CI/CD pipeline management, production deployment, AWS environment configuration, AWS Bedrock credential provisioning, and the core Role-Based Access Control (RBAC) framework. Team D applies the existing access control patterns to all new endpoints in this document but does not own or modify the underlying infrastructure.

## 7. Data Model Specification

### 7.1 Data Modeling Approach and Assumptions

CareConnect utilizes a normalized relational data model hosted on PostgreSQL, integrated via Spring Data JPA. All entities utilize UUID primary keys to prevent enumeration attacks. Soft-deletion is strictly enforced (records are marked `is\_deleted = true`, never dropped). Standard audit columns (created\_at, updated\_at, created\_by, updated\_by) exist on all tables. Schema migrations are managed via Flyway.

### 7.2 Communication Data Models

The Communication domain underpins the secure messaging, telehealth, and notification capabilities described in Section 6.2. All message bodies and attachment metadata are classified as PHI and are therefore encrypted at rest via AWS KMS-managed AES-256 keys; attachment binaries are stored in a dedicated S3 bucket with server-side encryption (SSE-KMS) and bucket policies that deny non-TLS access. Real-time WebSocket sessions are bound to a Conversation aggregate so that delivery, read-receipts, and audit events can be reconstructed deterministically.

| Entity       | Primary Key  | Key Attributes                                                                                    | Relationships                                   | Notes                                                       |
|--------------|--------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------|
| conversation | id :<br>UUID | care_circle_id, topic,<br>type (direct / group /<br>telehealth),<br>last_activity_at              | 1:M to message;<br>1:M to<br>participant        | Scoped by<br>care-circle<br>RBAC.                           |
| participant  | id :<br>UUID | FK conversation_id,<br>FK user_id,<br>role_in_conversation,<br>joined_at, left_at                 | M:1 to<br>conversation;<br>M:1 to user          | Soft-deleted;<br>tombstoned<br>for audit<br>replay.         |
| message      | id :<br>UUID | FK conversation_id, FK<br>sender_id,<br>body_ciphertext, sent_at,<br>delivered_at, read_at        | M:1 to<br>conversation;<br>1:M to<br>attachment | Body stored<br>ciphertext-only;<br>KMS data key<br>per row. |
| attachment   | id :<br>UUID | FK message_id,<br>s3_object_key, mime_type,<br>byte_size,<br>checksum_sha256                      | M:1 to message                                  | S3 SSE-KMS; pre-<br>signed URLs only.                       |
| call_session | id :<br>UUID | FK conversation_id,<br>chime_meeting_id,<br>started_at, ended_at,<br>session_type (audio / video) | M:1 to<br>conversation                          | Chime meeting<br>metadata only; no<br>media stored.         |

**Design notes:** Message search uses a separate inverted-index view populated from decrypted payloads only inside the application layer — the database never indexes plaintext PHI. Retention follows the HIPAA Designated Record Set policy (minimum 6 years) and is enforced by a scheduled archival job rather than hard deletes.

### 7.3 Electronic Visit Verification (EVV) Entities

The EVV domain implements the 21st Century Cures Act §12006 lifecycle and underpins Section 6.3. Visits transition through a strict state machine — *scheduled* → *in-progress* → *completed* → *submitted* → *exported* — and each transition is captured as an immutable `visit_event` row. Because caregivers routinely operate in low-connectivity environments, the model is designed for an **offline-first** client: events are queued locally in `offline_sync_queue` mirrors and reconciled server-side using deterministic UUIDs to guarantee idempotent replay within the SRS-mandated 24-hour offline window (SRS §2.4.3 / §6.4).

| Entity                          | Primary Key               | Key Attributes                                                                                                                                                                                                                                  | Relationships                                                             | Notes                                          |
|---------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------|
| <code>visit</code>              | <code>id</code> :<br>UUID | <code>scheduled_time</code> , <code>status</code> , FK<br><code>caregiver_id</code> , FK<br><code>patient_id</code> ,<br><code>submission_state</code> ,<br><code>service_code</code>                                                           | 1:M to <code>visit_event</code> ;<br>M:1 to <code>edi_export_batch</code> | State machine enforced at service layer.       |
| <code>visit_event</code>        | <code>id</code> :<br>UUID | FK <code>visit_id</code> , <code>event_type</code><br>( <code>check_in</code> / <code>check_out</code> / <code>pause</code> / <code>resume</code> ), <code>event_time</code> ,<br><code>device_id</code> , FK<br><code>geo_coordinate_id</code> | M:1 to <code>visit</code> ; 1:1 to <code>geo_coordinate</code>            | Append-only; never updated.                    |
| <code>geo_coordinate</code>     | <code>id</code> :<br>UUID | <code>latitude</code> , <code>longitude</code> ,<br><code>accuracy_m</code> , <code>captured_at</code> ,<br><code>source</code> ( <code>gps</code> / <code>network</code> )                                                                     | 1:1 with <code>visit_event</code>                                         | Geofence validation server-side;<br>REQ-1402 . |
| <code>offline_sync_queue</code> | <code>id</code> :<br>UUID | FK <code>visit_id</code><br>(nullable),<br><code>payload_json</code> ,<br><code>client_generated_at</code> ,<br><code>synced_at</code> ,<br><code>retry_count</code>                                                                            | M:1 to <code>visit</code>                                                 | 24-hour TTL; forced re-auth on expiry.         |

| Entity           | Primary Key  | Key Attributes                                                                                      | Relationships  | Notes                                         |
|------------------|--------------|-----------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------|
| edi_export_batch | id :<br>UUID | batch_status,<br>submitted_at,<br>aggregator_ref,<br>edi_transaction_set,<br>record_count, checksum | 1:M from visit | Batch immutable once submitted;<br>REQ-1404 . |

**Design notes:** The composite index (caregiver\_id, scheduled\_time) supports the most common caregiver-day query path; a partial index on status = 'submitted' accelerates the pre-export validator. The export batch checksum is a SHA-256 of the canonicalized payload, allowing the aggregator to detect tampering or partial transmission.

## 7.4 In-Home Residential Support Entities

This domain models the day-to-day caregiver workflow described in Section 6.4: per-resident care plans, task execution, shift handoffs, and incident reporting. Care plans are versioned rather than mutated — every edit produces a new effective\_range-bounded row so that historical task logs can always be reconciled against the plan that was active at execution time. This pattern is essential for audit defensibility under the HIPAA Designated Record Set.

| Entity           | Primary Key  | Key Attributes                                                                                              | Relationships                           | Notes                                          |
|------------------|--------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------|------------------------------------------------|
| resident_profile | id :<br>UUID | FK patient_id,<br>diagnoses, allergies,<br>preferences,<br>emergency_contact                                | 1:1 to user (patient); 1:M to care_plan | PHI-classified; column-level encryption.       |
| care_plan        | id :<br>UUID | FK resident_profile_id,<br>goals, interventions,<br>effective_range (tstzrange),<br>version, FK authored_by | 1:M to task; M:1 to resident_profile    | Versioned; never updated in place.             |
| task             | id :<br>UUID | FK care_plan_id,<br>description, frequency,<br>category, required (bool)                                    | M:1 to care_plan; 1:M to shift_log      | Drives caregiver daily checklist.              |
| shift_log        | id :<br>UUID | FK task_id, FK performed_by,<br>status (done / skipped / refused),                                          | M:1 to task; M:1 to user                | Notes screened by PHI redaction before export. |

| Entity          | Primary Key  | Key Attributes                                                                                                               | Relationships              | Notes                                                                          |
|-----------------|--------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------|
|                 |              | performed_at, notes, handoff_summary                                                                                         |                            |                                                                                |
| incident_report | id :<br>UUID | FK<br>resident_profile_id,<br>FK reported_by,<br>severity, category,<br>narrative_ciphertext,<br>occurred_at,<br>reported_at | M:1 to<br>resident_profile | Auto-generates<br>audit_event;<br>supervisor escalation<br>on severity ≥ HIGH. |

**Design notes:** The `effective_range` column uses PostgreSQL's native `tstzrange` type with an exclusion constraint preventing overlapping active plans per resident. Shift handoff summaries are materialized into a daily read-replica view to keep the primary write path lean.

## 7.5 AI-Assisted Services Entities

This domain backs the AI-assisted invoicing pipeline (Section 6.5) and the AWS Bedrock / Claude migration described in through. The schema is intentionally split into a business record half (`invoice`, `invoice_line_item`) and a model-interaction half (`extraction_request`, `extraction_result`, `model_prompt_log`) so that AI confidence, latency, and prompt lineage can be audited independently from the approved invoice itself. The `model_prompt_log` table stores redacted-only prompt and response text — raw PHI is never persisted to this surface, in conformance with the AWS BAA scope.

| Entity  | Primary Key  | Key Attributes                                                                                                                                | Relationships                                                | Notes                                                                                             |
|---------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| invoice | id :<br>UUID | vendor_name,<br>invoice_number,<br>invoice_date,<br>total_amount,<br>currency, status,<br>confidence_score, FK<br>approved_by,<br>approved_at | 1:M to<br>invoice_line_item;<br>1:M to<br>extraction_request | Status<br>cannot<br>advance<br>to<br>accepted without<br>human<br>approver<br>(<br>REQ-1507<br>). |

| Entity                  | Primary Key  | Key Attributes                                                                                                                                     | Relationships                                                              | Notes                                                                                     |
|-------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| invoice_line_item       | id :<br>UUID | FK invoice_id,<br>description, qty,<br>unit_price,<br>line_total,<br>field_confidence_j<br>son                                                     | M:1 to invoice                                                             | Per-field confidenc<br>e drives<br>UI badge<br>color.                                     |
| extraction_reques<br>t  | id :<br>UUID | FK invoice_id, model<br>(e.g.,<br>claude-3.5-sonnet@b<br>edrock),<br>submitted_at,<br>completed_at,<br>latency_ms,<br>token_input,<br>token_output | M:1 to invoice; 1:1 to<br>extraction_result;<br>1:1 to<br>model_prompt_log | Retries<br>produce<br>new rows<br>— never<br>updated.                                     |
| extraction_result       | id :<br>UUID | FK<br>extraction_reques<br>t_id,<br>extracted_fields_<br>json,<br>overall_confidenc<br>e, warnings_json                                            | 1:1 to<br>extraction_request                                               | Source of<br>pre-fill<br>values for<br>reviewer<br>UI (<br><a href="#">REQ-1503</a><br>). |
| reviewer_assignme<br>nt | id :<br>UUID | FK invoice_id, FK<br>reviewer_id,<br>assigned_at,<br>completed_at,<br>decision (approve<br>/ reject / escalate)                                    | M:1 to invoice; M:1 to<br>user                                             | Enforces<br>dual-contr<br>ol on<br>totals<br>above<br>policy<br>threshold.                |
| model_prompt_log        | id :<br>UUID | FK<br>extraction_reques<br>t_id,<br>redacted_prompt,<br>redacted_response<br>,<br>phi_filter_versio<br>n, created_at                               | 1:1 to<br>extraction_request                                               | <b>Append-o<br/>nly.</b><br>Redacted<br>text only;<br>never<br>raw PHI.                   |

**Design notes:** Confidence scores propagate from extraction\_result.overall\_confidence down to

invoice\_line\_item.field\_confidence\_json so the Flutter reviewer UI can render per-field badges. A database trigger blocks any UPDATE/DELETE against model\_prompt\_log, providing defense-in-depth on top of the application-level append-only contract.

## 7.6 Administration & Audit Entities

The Admin & Audit domain provides identity, authorization, and the immutable evidentiary trail required for HIPAA

§164.308 / §164.312 compliance. The audit\_event table is the single source of truth for PHI access, configuration change, and security-relevant operations; it is append-only at three layers — application, ORM listener, and database trigger — so that even a compromised service account cannot rewrite history. Wearable telemetry (wearable\_data\_point) lives in this domain because it shares the same per-patient access-control surface as identity and audit.

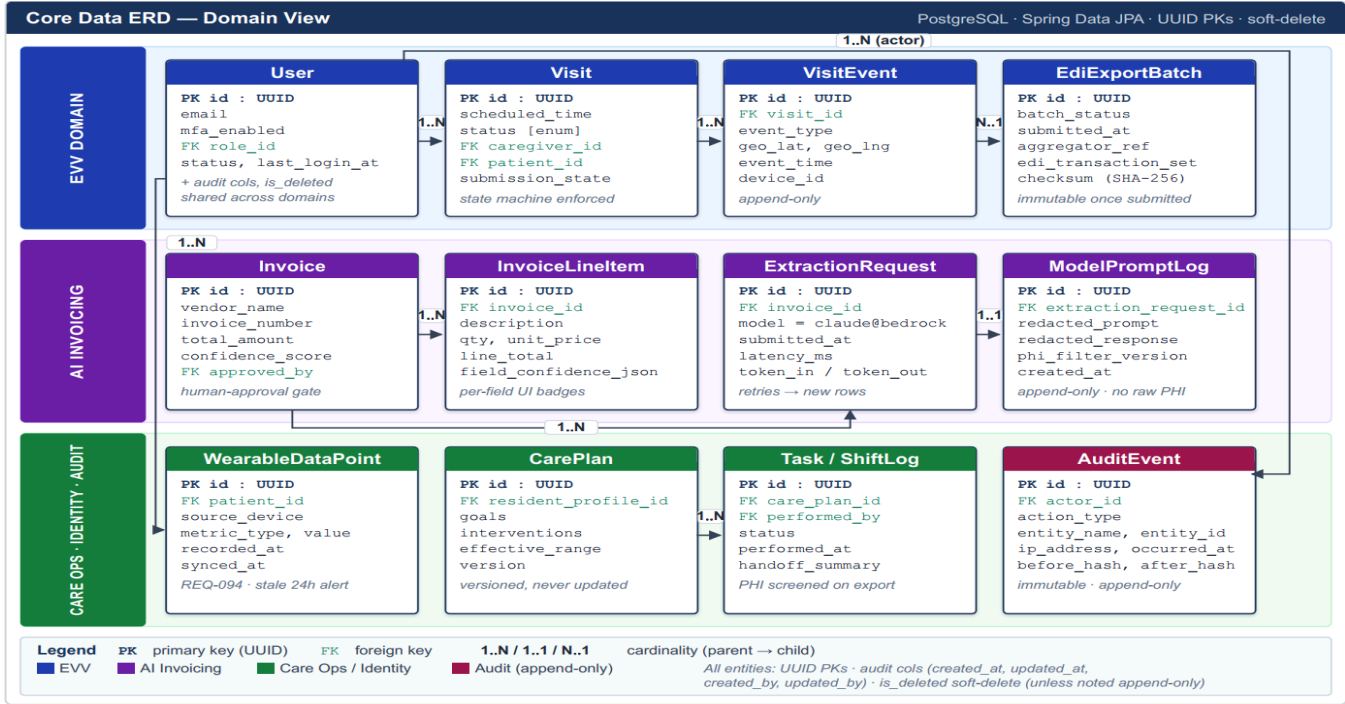
| Entity      | Primary Key  | Key Attributes                                                                                                                                            | Relationships                                  | Notes                                                   |
|-------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------|
| user        | id :<br>UUID | email, mfa_enabled,<br>password_hash (Argon2id),<br>status, last_login_at,<br>FK role_id                                                                  | M:1 to role; 1:M<br>to<br>audit_event          | Email unique; MFA mandatory for clinical roles.         |
| role        | id :<br>UUID | name (caregiver / supervisor<br>/ admin / reviewer / patient),<br>description                                                                             | M:N to<br>permission<br>via<br>role_permission | Seeded by Flyway; not user-editable.                    |
| permission  | id :<br>UUID | code (e.g.,<br>VISIT.SUBMIT,<br>INVOICE.APPROVE),<br>description                                                                                          | M:N to role                                    | Evaluated by Spring Security<br>@PreAuthorize.          |
| audit_event | id :<br>UUID | FK actor_id,<br>action_type,<br>entity_name,<br>entity_id,<br>ip_address,<br>user_agent,<br>correlation_id,<br>occurred_at,<br>before_hash,<br>after_hash | M:1 to user<br>(actor)                         | <b>Append-only;</b> DB trigger<br>blocks UPDATE/DELETE. |

| Entity              | Primary Key  | Key Attributes                                                                                                                            | Relationships            | Notes                                                                        |
|---------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------|
| compliance_report   | id :<br>UUID | report_type<br>(access_review /<br>phi_disclosure /<br>breach_drill),<br>period_start,<br>period_end,<br>s3_evidence_key,<br>generated_at | N/A (snapshot artifact)  | Generated on read-replica to avoid primary contention.                       |
| wearable_data_point | id :<br>UUID | FK patient_id,<br>source_device,<br>metric_type, value,<br>unit, recorded_at,<br>synced_at                                                | M:1 to user<br>(patient) | Lineage retained per<br>REQ-094 ; stale-sync alert at 24h<br>(<br>REQ-095 ). |

**Design notes:** Each `audit_event` stores SHA-256 hashes of the before/after JSON snapshots rather than the payloads themselves, keeping the table lean while still permitting tamper detection. The role/permission join table is cached in Redis on application startup and invalidated via a Postgres `NOTIFY` channel, so authorization checks remain sub-millisecond even at peak load.

### 7.7 Data Dictionary Alignment & ER Structure

The system maintains a traceability matrix linking SRS REQ-IDs directly to JPA entities to align with the HIPAA Designated Record Set.





| Entity Table | Primary Key (UUID) | Key Attributes                                            | Relationships            | Audit / Soft Delete |
|--------------|--------------------|-----------------------------------------------------------|--------------------------|---------------------|
| visit        | id                 | status, scheduled_time, caregiver_id, patient_id          | 1:M to visit_event       | Yes                 |
| invoice      | id                 | vendor_name, total_amount, confidence_score, status       | 1:M to invoice_line_item | Yes                 |
| audit_event  | id                 | action_type, entity_name, entity_id, actor_id, ip_address | N/A (Immutable Log)      | No (Append-Only)    |

## 8. UI / UX Design Specification

Team D UI design keeps screens consistent with the shared CareConnect design direction. The approach prioritizes clear labels and non-technical user messages, accessibility and usability before visual polish, and explicit identification of all features that change navigation, forms, messages, or workflows. The six Team D feature modules — EVV, AI-Assisted Invoicing, Onboarding, Wearable Integration, Document Digitization, and Screening/Assessment — each introduce new or modified screens that are designed to minimize cognitive load and preserve workflow continuity.

### 8.1 UI Design Approach and Principles

- Keep screens consistent with the shared CareConnect design direction.
- Use clear labels and non-technical user messages.
- Design for accessibility and usability before visual polish.
- Identify any feature that changes navigation, forms, messages, or workflows.
- Write Not applicable if the assigned work does not affect UI or UX.

### 8.2 Screens and User Workflows Affected

| Screen / Workflow             | Change or Design Need                                                                             | Affected Users               | Dependencies                        |
|-------------------------------|---------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------|
| EVV Dashboard                 | Add aggregator readiness status indicators and submission preparation visibility.                 | Caregivers, Admins           | EVV aggregator status API           |
| EVV Visit Detail              | Add correction and retry interaction states for invalid or incomplete records.                    | Caregivers, Admins           | EVV Write Endpoints                 |
| Invoice Review                | Add manual review fallback path and failure messaging for low-confidence or failed extractions.   | Billing users, Reviewers     | Invoice status API                  |
| Invite/Onboarding Entry       | Add token intake screen, inviter context display, and token error states (invalid, expired, used) | New users, Inviters          | Invite token API, deep-link routing |
| Registration and Role Routing | Add verification checkpoints, role assignment confirmation, and role-specific dashboard routing.  | New users                    | Auth and role endpoints             |
| Wearable Connect              | Add connector authorization flow, link status, and connection health display.                     | Patients, Caregivers         | Wearable connector OAuth adapters   |
| Wearable Dashboard            | Add latest sync timestamp, metric display, threshold indicators, and acknowledgement actions.     | Patients, Caregivers, Admins | Wearable metric ingestion service   |
| Home Care Document Form       | Add structural digital form with field validation, draft save, and submit workflow                | Caregivers, Patients         | Document Storage API                |
| Document Review               | Add admin accept/return actions and correction guidance display.                                  | Admins, Reviewers            | Document status API                 |
| Screening From                | Add structured multi-care field screening capture with draft/submit lifecycle.                    | Assessors                    | Assessment API                      |
| Comprehensive Assessment      | Add multi-section form with domain grouping, draft resume, risk/disposition                       | Assessors                    | Assessment and patient profile API  |

|  |                                  |  |  |
|--|----------------------------------|--|--|
|  | capture, and escalation prompts. |  |  |
|--|----------------------------------|--|--|

### 8.3 Wireframes, Mockups, or Screen Notes

#### 8.3.1 EVV Dashboard

1. EVV Enhancements + Aggregator Dependency Tracking

Screen: EVV Operations Dashboard

Home

EVV Dashboard

Visit In Progress

Corrections Queue

Offline Sync

Aggregator Dependencies

Reports

Open Visits

18

Sync Failures

4

Dependency Alerts

2

FILTERS

Caregiver

Date Range

Status

Apply

Export CSV

| Visit ID | Patient   | Check-In/Out  | Status            | Action  |
|----------|-----------|---------------|-------------------|---------|
| EVV-1902 | M. Rivera | 09:05 / -     | In Progress       | View    |
| EVV-1901 | T. Brown  | 08:00 / 08:47 | Pending Sync      | Retry   |
| EVV-1897 | L. Jones  | 07:30 / 08:15 | Correction Needed | Resolve |

Start Visit

Check-In Location

Visit In Progress

Check-Out + Sync

Dependency Panel: Shows external EVV aggregator state, stale webhooks, and mapping mismatches before finalizing records.

- Aggregator health heartbeat
- Queue backlog count
- Manual replay control
- Audit trail for correction edits

Dashboard uses card and table layout to surface visit state, aggregator readiness, and submission status. Detail screens show correction and retry affordances inline without full-page navigation disruption.

8.3.2 AI Invoicing

2. AI-Assisted Invoicing with Bedrock/Claude

Screen: Invoice Assistant Workbench

Dashboard

Invoice Assistant

Upload Batch

Review Queue

Approvals

History

INTAKE

Upload PDF/Image

Capture from Scanner

Model: Claude (via Bedrock)

DOCUMENT PREVIEW

AI EXTRACTION PANEL

Invoice #

INV-4582

Provider

CareConnect Agency

Amount

\$324.00

Accept AI Fields

Flag for Manual Review

| Rule Check        | Status  | Reason           | Confidence | Override |
|-------------------|---------|------------------|------------|----------|
| Duplicate Invoice | Warning | Similar # exists | 84%        | Review   |
| Missing CPT Code  | Fail    | Line 2 missing   | 92%        | Edit     |

Human-in-the-loop flow:  
Upload -> AI parse -> policy checks -> reviewer approval -> submit.

- Save model response metadata
- Expose confidence and rationale
- Track overrides for QA
- Support retrievable extraction

Intake screen preserves the existing upload interaction. Extraction result view presents structured fields in an editable form with confidence indicators visible before approval is possible. Manual review routing is explicit and always available.

8.3.3 QR Onboarding

3. QR Code + Invite-Link Onboarding

Screen: Onboarding Entry + Verification

Onboarding

Invite Management

QR Credentials

Access Log

INVITE LINK ENTRY

https://careconnect.app/invite/abc123

Validate Link

ACCOUNT BASICS

First Name

Last Name

Role

Continue

SCAN QR

Use Webcam

Upload QR Image

| Step               | Status  | Input       | Output          | Next    |
|--------------------|---------|-------------|-----------------|---------|
| Token Validation   | Pass    | Invite Link | Pre-filled user | KYC     |
| Email Verification | Pending | OTP         | Verified        | Consent |
| Role Confirmation  | Pending | Role Select | Scoped access   | Finish  |

Unified onboarding endpoint supports both deep link and QR payloads with the same verification pipeline.

- Invite expiration and revocation checks
- Single-use token handling
- Desktop QR camera fallback options
- Post-onboarding redirect by role

Token intake screen shows inviter context and accept/reject controls. Error states for expired, invalid, and already-used tokens each have distinct non-technical messaging and a next-step prompt.

8.3.4 Wearables Integration

4. Wearable Device Integration

Screen: Device Hub + Data Sync

Health Home

Wearables

Connected Devices

Sync Logs

Alerts

Devices Connected

3

Last Sync

2 min ago

Signal Quality

Good

ADD DEVICE

Connect Fitbit

Connect Apple Health

Connect Google Health

| Metric     | Latest  | Source       | Timestamp | Status  |
|------------|---------|--------------|-----------|---------|
| Heart Rate | 76 bpm  | Fitbit Versa | 10:41 AM  | Synced  |
| Steps      | 4,220   | Fitbit Versa | 10:39 AM  | Synced  |
| Sleep      | 6.8 hrs | Apple Health | 07:10 AM  | Partial |

Feasibility-focused design:  
explicit connector states,  
permission prompts, and sync  
observability for staged  
rollout.

- Token lifecycle and re-auth reminders
- Normalize units before storage
- Mark stale metrics clearly
- Surface API quota and errors

Connect screen walks through the provider authorization flow step by step. Dashboard panels show sync timestamp recency and threshold alert status with clear acknowledgment controls.

8.3.5 Document Digitization

5. Home Care Document Digitization

Screen: Document Intake + Structured Output

Patient Record

Documents

OCR Queue

Validation

Exports

CAPTURE METHODS

Upload PDF

Upload Image

Drag and Drop Batch

Categories: Visit Notes, Insurance, Medication, Consent

DOCUMENT VIEWER

EXTRACTED FIELDS

Patient Name: Maria Rivera

Visit Date: 2026-05-24

Provider Signature: Detected

Confirm and Save

| Doc ID  | Type       | OCR Status     | Validator | Action |
|---------|------------|----------------|-----------|--------|
| DOC-991 | Visit Note | Parsed         | Pending   | Review |
| DOC-990 | Insurance  | Low Confidence | Assigned  | Edit   |

Digitization pipeline supports  
confidence scoring, manual  
correction, and auditable  
version history.

- Retain original file and parsed JSON
- Highlight low-confidence fields
- Track user corrections
- Export clean records to care plan workflows

Forms use a section-by-section layout with visible required field indicators. Draft and submit actions are clearly differentiated. Review status is surfaced prominently on the document history screen.

### 8.3.6 Screening and Assessment

6. Screening and Assessment Workflows

Screen: Assessment Builder + Completion

Clinical Home

Assessments

Templates

Assigned

Completed

ASSESSMENT HEADER

Patient: T. Brown

Template: Fall Risk Screen

Assessor: Caregiver 204

Start Session

QUESTION FLOW

| # | Question                   | Response | Risk Weight | Notes         |
|---|----------------------------|----------|-------------|---------------|
| 1 | Any falls in last 30 days? | Yes      | High        | Two incidents |
| 2 | Uses mobility aid?         | No       | Medium      | -             |
| 3 | Dizziness episodes?        | Yes      | High        | Morning only  |

Current Risk Score

72 / 100

DECISION

Save Draft

Submit Assessment

Escalate Alert

Supports template-driven screening with scoring logic and escalation paths to follow-up workflows.

- Auto-save draft responses
- Role-based visibility and editing
- Scoring rubric version control
- Post-submit action orchestration

Multi-section forms use progressive disclosure per domain. Draft save and resume are available at any point. Risk and disposition capture includes an explicit confirmation step before final submission is allowed.

### 8.4 Accessibility Requirements

| Accessibility Area    | Design Requirement                                                                                                                      |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Keyboard navigation   | Users should be able to complete key workflows without a mouse when applicable.                                                         |
| Screen reader support | Interactive controls should have clear labels when applicable.                                                                          |
| Color contrast        | Text and key UI elements should meet readable contrast expectations.                                                                    |
| Responsive behavior   | Supported screen sizes and orientations should remain usable.                                                                           |
| Error messages        | Messages should be clear, specific, and non-technical.                                                                                  |
| Touch targets         | Interactive controls should meet minimum touch target sizes for mobile use (at least 44x44 px; preferred 48x48 px for primary actions). |

### 8.5 Display Text and Localization Impact

| Screen / Area           | UI Text or Text Key                                                                        | Needs Translation? | Notes                                    |
|-------------------------|--------------------------------------------------------------------------------------------|--------------------|------------------------------------------|
| EVV Dashboard           | Aggregator status labels, submission readiness indicators, state transition labels         | No                 | English only for current milestone scope |
| Invoice Intake          | Upload prompt, extraction status messages, confidence warning labels, manual review prompt | No                 | English only for current milestone scope |
| Onboarding Token Screen | Invite acceptance prompt, error messages for invalid/expired/used                          | No                 | English only for current milestone scope |

|                      |                                                                                                      |    |                                          |
|----------------------|------------------------------------------------------------------------------------------------------|----|------------------------------------------|
|                      | tokens, role assignment confirmation                                                                 |    |                                          |
| Wearable Connect     | Authorization step labels, link status messages, sync freshness warning, alert acknowledgment prompt | No | English only for current milestone scope |
| Document Forms       | Field labels, required field indicators, draft/submit action labels, review status messages          | No | English only for current milestone scope |
| Screening/Assessment | Section headings, required field guidance, risk/disposition labels, escalation prompt                | No | English only for current milestone scope |

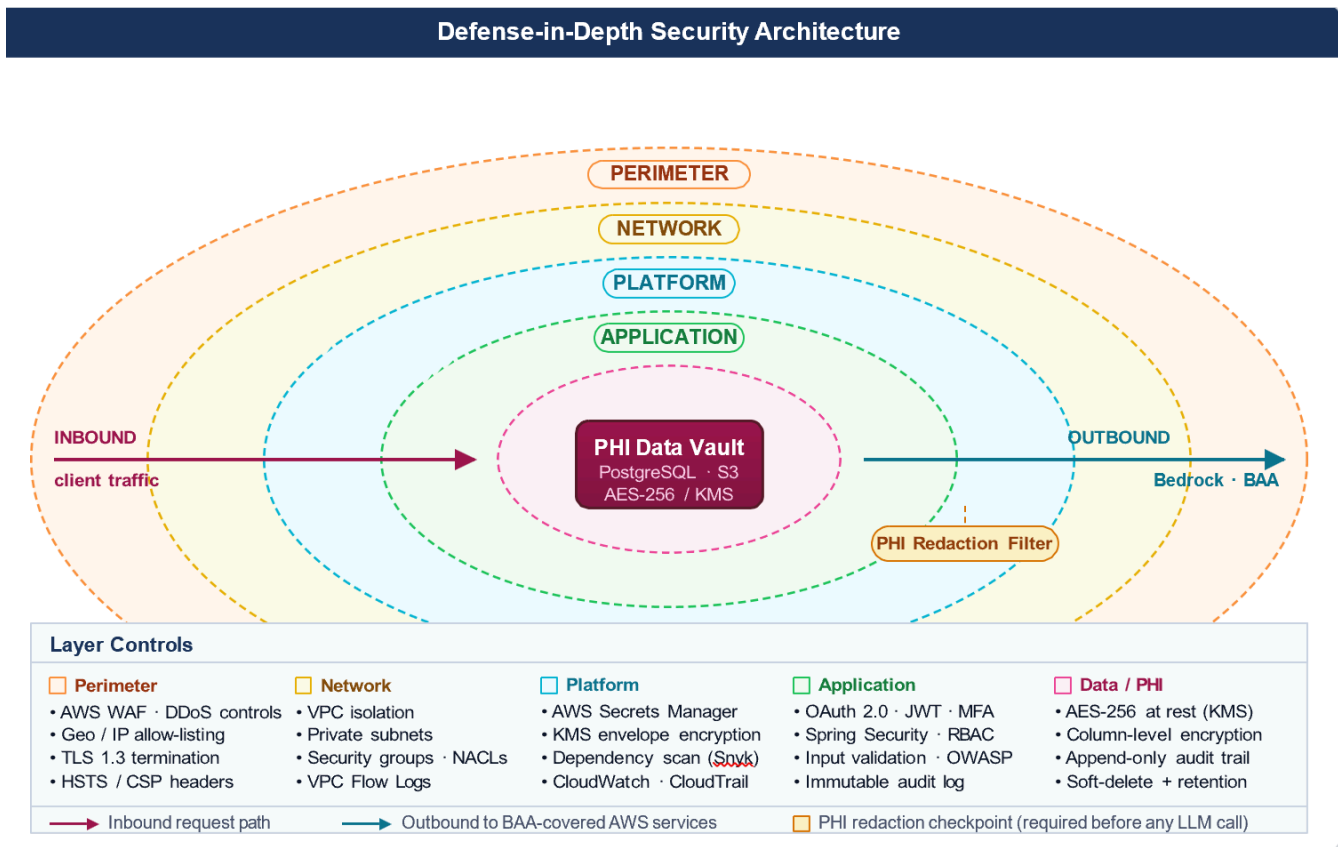
# 9. Security Considerations

## 9.1 Security Controls

CareConnect follows a defense-in-depth security model to ensure the confidentiality, integrity, and availability of all data, specifically PHI.

### HIPAA ALIGNMENT

All infrastructure conforms to the HIPAA Security Rule. A Business Associate Agreement (BAA) is established with AWS. Per Project Plan v2.0 §8.2 (Risk R-003 mitigation), the team will “*apply pre-prompt redaction for all PHI/PII fields*” and “*operate Bedrock only under BAA-covered AWS configuration.*” The PHI redaction middleware described in Section 6.5 and the redacted-only model\_prompt\_log persistence in Section 7.5 are the concrete implementations of this control.



**Figure 3.** Defense-in-depth security architecture. Inbound traffic crosses successive perimeter, network, platform, and application boundaries before reaching the PHI data vault; outbound calls to AWS Bedrock pass through a dedicated PHI redaction filter under BAA scope.



| Control Category                | Implemented Mechanism                                                                                             | Standard / Source                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <b>Auth &amp; Authorization</b> | OAuth 2.0 / JWT (HS256 or RS256), MFA, RBAC via Spring Security filter chain.                                     | SRS Onboarding & Auth module; least privilege.                        |
| <b>Encryption</b>               | TLS 1.3 in transit; AES-256 at rest via AWS KMS; envelope encryption for S3.                                      | SRS §6.1 Data Encryption.                                             |
| <b>Network Infrastructure</b>   | AWS WAF at the public edge; managed API entry points; DB in RDS managed environment.                              | Project Plan §Security; SRS Cloud Hosting.                            |
| <b>Application Security</b>     | Input validation, CSRF tokens, HSTS/CSP headers, <b>Snyk</b> dependency scanning in CI.                           | OWASP Top 10 mitigations against injection & XSS.                     |
| <b>Logging &amp; Audit</b>      | Immutable audit logs for PHI access; AWS CloudWatch metrics; anomaly alerts.                                      | SRS Audit requirements; HIPAA §164.316(b)(2) documentation retention. |
| <b>AI / PHI Boundary</b>        | PHI redaction middleware before AWS Bedrock invocations; redacted-only <code>model_prompt_log</code> persistence. | SRS §HIPAA §164.308(b)(1) — BAA-covered LLM data exposure surface.    |
| <b>Data Integrity</b>           | Server-side validation; append-only audit log; signed exports.                                                    | HIPAA §164.312(c)(1) integrity controls (cited in SRS).               |
| <b>Secrets Management</b>       | AWS Secrets Manager with automated rotation; no hard-coded keys.                                                  | SRS Cloud Hosting; Project Plan secret-store guidance.                |

## 9.1 Data Privacy

CareConnect handles sensitive healthcare information and is designed with privacy protection as a core requirement. Access to patient information is restricted to authorized users based on their role and relationship to the individual receiving care. System activity involving protected data is recorded to support accountability, auditing, and compliance reviews. Data management practices are intended to limit unnecessary exposure, replication, or sharing of healthcare information.

## 9.2 Authentication and Authorization

Access to CareConnect functionality requires successful user authentication through secure identity verification mechanisms. Once authenticated, permissions are granted according to assigned roles and applicable care relationships. The platform enforces access restrictions at both the application and service layers to ensure users can only perform actions appropriate to their responsibilities. Failed login attempts and unauthorized access events are recorded for security monitoring purposes.

## 9.3 Secure Data Transmission

Information exchanged between users, services, and external integrations is protected through industry-standard encryption protocols. Secure communication channels are used for both traditional API traffic and real-time

system interactions. These safeguards help maintain confidentiality and prevent unauthorized interception or modification of data while in transit.

#### **9.4 API Security**

All exposed APIs require appropriate authentication and authorization checks before processing requests. Incoming data is validated to reduce the risk of malicious input, improper formatting, or injection-based attacks. Additional protections such as request throttling, payload size restrictions, and centralized error handling help safeguard the platform against abuse and service disruptions.

#### **9.5 Data Storage and Backup**

CareConnect stores operational and healthcare-related information using secure storage mechanisms that support encryption and controlled access. Routine backup procedures are implemented to support disaster recovery and business continuity objectives. Backup data is maintained separately from active production environments to reduce the risk of data loss resulting from system failures or security incidents.

#### **9.6 Logging and Monitoring**

The mobile application follows security best practices to protect user sessions and locally stored information. Authentication tokens and other sensitive session data are stored using platform-supported secure storage solutions. Automatic session expiration and inactivity controls help reduce the likelihood of unauthorized access if a device is lost, stolen, or left unattended.

#### **9.7 External Service or AI Tool Security**

AI-enabled functionality within CareConnect is delivered through AWS Bedrock and operates under the same security controls applied to other backend services. Prior to processing, submitted information is reviewed and minimized to limit unnecessary exposure of sensitive healthcare data. Communications between CareConnect and AI service providers occur through encrypted connections, and system activity is logged to support traceability and auditing. AI-generated outputs require user review and confirmation before becoming part of any permanent record.

#### **9.8 Security Testing**

Emergency and high-priority notification capabilities are designed to maximize reliability during critical situations. The system incorporates fault-tolerant communication mechanisms and alternative delivery paths when primary communication channels are unavailable. These safeguards help ensure that urgent alerts and notifications continue to reach intended recipients even during periods of degraded connectivity or service interruptions.

## 10. Error Handling

To prevent sensitive data leakage while providing actionable debugging information, CareConnect utilizes a globally standardized error response envelope. The global Spring `@ControllerAdvice` exception handler intercepts all thrown exceptions and sanitizes them before reaching the Flutter client.

- **HTTP Status Codes:** 400 (Bad Request), 401 (Unauthorized), 403 (Forbidden), 404 (Not Found), 409 (Conflict), 422 (Unprocessable Entity), 429 (Too Many Requests), 500 (Internal Server Error), 503 (Service Unavailable).
- **Resiliency:** External dependencies (AWS Bedrock, Chime, wearable APIs) are wrapped using the Resilience4j circuit-breaker library. Transient failures trigger retry/backoff with jitter.
- **Offline Error Queuing:** EVV actions occurring during network disruption are locally queued in the Flutter client within the SRS-mandated **24-hour offline window** (SRS §2.4.3 / §6.4) and synchronized later. After 24 hours, forced re-authentication is enforced per SRS.

### SECURITY NOTE

Internal stack traces and PHI are NEVER serialized into the `details` or `message` fields of an API error response.

```
{
  "timestamp":
    "2026-06-12T14:32:01.123Z", "traceId":
    "1-67891234-abcdef1234567890", "code":
    "ERR_409_CONFLICT",
  "message": "The EVV Check-in record conflicts with an existing entry.",
  "details": [
    "Visit ID already has an active check-in timestamp."
  ]
}
```

## 11. Performance & Scalability

### 11.1 Objectives

Performance objectives are anchored to SRS §2.4.3 (Network Requirements) and §2.5.3 (Throughput Constraints). All performance targets are measured at the **95th percentile under production-like load**, per the SRS Performance & Scalability section.

| Operation Type                         | p95 Latency Target                              | Source                                                                                   |
|----------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------|
| Network Round-Trip (client ↔ API)      | ≤ 300 ms                                        | SRS §2.4.3 — mandatory ceiling                                                           |
| Standard Read APIs (cache hit)         | < 500 ms                                        | Approved baseline — includes Redis cache (SRS-confirmed component)                       |
| Standard Write APIs                    | < 800 ms                                        | Approved baseline — includes DB commit + audit-log write                                 |
| EVV Check-in/out submission            | < 700 ms (online) ·<br>queued (offline)         | 24-hour offline window per SRS §2.4.3 / §6.4                                             |
| AI Invoice Extraction (Bedrock/Claude) | < 1.5 s synchronous ·<br>async preferred > 1 MB | Bound by Bedrock TPS quota (SRS §2.5.3);<br><a href="#">REQ-1501</a>                     |
| Wearable telemetry ingest              | < 1.0 s per data point                          | Approved baseline; batched on device,<br><a href="#">REQ-094</a>                         |
| Availability (rolling 30 days)         | ≥ 99.5%                                         | Approved baseline; backup/recovery via RDS PITR with <b>RPO ≤ 15 min, RTO ≤ 1 hour</b> . |

### 11.2 Scalability Strategy

Per the Project Plan and SRS Cloud Hosting section, CareConnect uses a serverless-first architecture on AWS Lambda fronted by AWS API Gateway. Backend services are stateless to enable horizontal scaling under Lambda concurrency. Amazon RDS for PostgreSQL is the primary store, with read replicas dedicated to reporting workloads. Asynchronous workflows (AI invoice extraction, batch EVV sync) are decoupled via AWS SQS.

### 11.3 Performance Monitoring

AWS CloudWatch (SRS Cloud Hosting) is configured with custom dashboards for system latency, error rates, and throughput. AWS X-Ray provides distributed tracing across services, and Spring Boot Actuator metrics expose JVM and connection-pool health (HikariCP, per SRS).

---

### 11.4 Bottleneck Mitigation

To sustain performance, specific mitigations include:

- Database Optimization: Comprehensive query indexing; connection pooling via HikariCP (explicitly named in SRS).
- Caching: Redis Cache (SRS-confirmed active component) is used to reduce DB load for frequently read data (e.g., role definitions, unchanged care plans).
- Payload Processing: Uploaded images are compressed prior to AWS Textract OCR; EDI exports are batched; Bedrock prompts are dynamically size-optimized to prevent token-limit exhaustion.

### Additional Responsibilities

Beyond the core TDD sections, the QA / Security Lead assumes oversight for the following continuous lifecycle tasks. Per Project Plan v2.0 §3.2.9, Team A owns the CI/CD pipeline platform itself; Team D supplies the test artifacts and security gates that run on it. Per the v2.0 §10.1 roles table, the QA / Security role is *Responsible* for “testing workflows, regression testing, defect verification” and is *Responsible* for Testing & QA and Documentation in the §10.2 RACI matrix.

- **Security Reviews:** Conducting per-sprint threat modeling exercises using the **STRIDE** methodology (Team D extended decision) for all newly designed features.
- Vulnerability Scanning (Project Plan v2.0 §6): Running dependency scanning, static analysis, and container/image scanning in CI before every release, plus targeted penetration-style test scenarios for high-risk workflows (EVV submission, AI invoice approval, PHI export).
- **Test Planning:** Authoring the Software Test Plan (Milestone 2 deliverable, due 06/13/2026 per Project Plan v2.0 §5) covering unit, integration, contract, End-to-End (E2E), security/penetration, and performance/load testing scenarios. A refreshed Software Test Plan is also a Milestone 3 deliverable (due 07/25/2026), bundled with the Programmer Guide and Deployment & Operations Guide.
- **Performance Validation:** Executing load-test baselines at the conclusion of each milestone to verify SLA adherence using **JMeter** (primary) and **k6** (CI smoke tests). Both tools are Team D extended decisions; Project Plan v2.0 does not name a specific load-testing tool.
- **Traceability Review:** Maintaining the SRS REQ-ID → TDD component → test-case mapping matrix to demonstrate full requirements coverage in the Test Report (Milestone 4, 08/04/2026 per Project Plan v2.0 §5).
- **Risk Register Alignment:** Reviewing test coverage against Project Plan v2.0 §8.2 risks — particularly (PHI/AI data leak), (token replay), and (EVV aggregator access) — to ensure each risk has at least one corresponding test case before each milestone gate.

## 12. Testing Strategy

### 12.1 Testing Overview

Testing activities for Team D will ensure that all assigned CareConnect enhancements function correctly, integrate successfully with existing platform components, and satisfy the requirements defined within the Software Requirements Specification (SRS). Testing will focus on validating Electronic Visit Verification (EVV) enhancements, AI-assisted invoicing workflows, QR code and invite-link onboarding, wearable integration functionality, home care document digitization, and assessment workflows.

The testing approach combines manual and automated testing techniques to verify functional behavior, usability, integration points, data integrity, security controls, and system reliability. Because several Team D features depend on external services that are not yet available for production use, testing will utilize mock implementations, test fixtures, and simulated service responses where necessary.

Testing activities will be conducted throughout development and will support milestone reviews, implementation validation, and final acceptance testing.

### 12.2 Testing Levels

#### 12.2.1 Unit Testing

Unit testing will validate individual components, services, utility functions, and business logic in isolation.

Focus areas include:

- EVV validation rules and workflow logic
- Invoice extraction processing and data mapping
- QR code generation and invitation token validation
- Wearable data synchronization logic
- Home care form validation rules
- Assessment workflow calculations and state management

Unit tests will be executed automatically through the project's continuous integration pipeline whenever code changes are submitted.

#### 12.2.2 Integration Testing

Integration testing will validate interactions between system components and external dependencies.

Focus areas include:

- Frontend-to-backend communication
- API request and response validation
- Database persistence operations
- Invoice workflow integration with AI service interfaces
- QR onboarding workflow integration with user management services
- Wearable data synchronization workflows
- EVV submission workflow readiness and payload generation

Where production services are unavailable, mock implementations and test fixtures will be used to validate integration behavior.

#### 12.2.3 System Testing

System testing will validate complete end-to-end workflows from the perspective of end users.

Key scenarios include:

- Creating and managing EVV records
- Uploading and reviewing invoices
- Joining a care circle through QR code or invite link
- Connecting wearable devices and viewing synchronized data
- Completing digital home care forms
- Completing screening and assessment workflows

System testing will verify that workflows function correctly across multiple user roles and device types.

#### 12.2.4 Security Testing

Security testing will verify that Team D features properly protect sensitive healthcare information and enforce access restrictions.

Focus areas include:

- Authentication and authorization validation
- Role-based access control (RBAC)
- Secure handling of invitation tokens
- PHI protection and data access restrictions
- Input validation and error handling
- Secure API communication

Security testing will also verify that mock implementations do not expose sensitive information during development and testing activities.

#### 12.2.5 Accessibility Testing

Accessibility testing will verify that Team D features are usable by individuals with diverse abilities and comply with applicable accessibility best practices. Testing will focus on ensuring that users can successfully interact with EVV workflows, invoice review screens, onboarding processes, wearable dashboards, and digital forms regardless of physical or cognitive limitations.

#### Testing Objectives

- Verify compliance with WCAG 2.1 Level AA accessibility guidelines where feasible.
- Ensure all functionality can be accessed using keyboard navigation.
- Validate compatibility with screen readers and assistive technologies.
- Confirm adequate color contrast and readability across all interfaces.
- Ensure form inputs include appropriate labels, instructions, and error messaging.

Verify responsive layouts remain accessible across desktop, tablet, and mobile devices.

### 12.3 Feature-Specific Test Cases

#### 12.3.1 EVV Test Coverage

- **EVV-001:** Caregiver completes valid check-in for assigned patient → **SUCCESS**
- **EVV-002:** Caregiver attempts check-in for unassigned patient → **BLOCKED**
- **EVV-003:** Caregiver completes valid check-out after active check-in → **SUCCESS**
- **EVV-004:** Caregiver attempts check-out without active check-in → **BLOCKED**
- **EVV-005:** EVV record contains missing required fields → **VALIDATION ERROR**

- **EVV-006:** Supervisor reviews EVV exception record → **SUCCESS**
- **EVV-007:** User without supervisor/admin role attempts EVV exception review → **BLOCKED**
- **EVV-008:** EVV payload preview is generated without live aggregator submission → **SUCCESS**
- **EVV-009:** System attempts live aggregator submission without approved credentials → **BLOCKED**
- **EVV-010:** EVV correction is submitted with reason and audit trail → **SUCCESS**

### 12.3.2 AI-Assisted Invoice Test Coverage

- **INV-001:** User uploads supported invoice file type → **SUCCESS**
- **INV-002:** User uploads unsupported invoice file type → **VALIDATION ERROR**
- **INV-003:** Invoice extraction runs using mock OCR/LLM fixture → **SUCCESS**
- **INV-004:** Extracted invoice fields display in reviewer workflow → **SUCCESS**
- **INV-005:** Reviewer edits extracted invoice fields before approval → **SUCCESS**
- **INV-006:** Reviewer approves invoice after human review → **SUCCESS**
- **INV-007:** System attempts to auto-approve invoice without human review → **BLOCKED**
- **INV-008:** Duplicate invoice is detected during review → **WARNING**
- **INV-009:** Bedrock/Claude provider is unavailable during extraction → **FALLBACK / MANUAL REVIEW**
- **INV-010:** User without billing/reviewer permissions accesses invoice review screen → **BLOCKED**

### 12.3.3 QR Code and Invite-Link Onboarding Test Coverage

- **QR-001:** Authorized user generates invite link → **SUCCESS**
- **QR-002:** Authorized user generates QR code invitation → **SUCCESS**
- **QR-003:** Invited user opens valid invite link → **SUCCESS**
- **QR-004:** Invited user scans valid QR code → **SUCCESS**
- **QR-005:** User accepts valid invitation and joins care circle → **SUCCESS**
- **QR-006:** Expired invitation token is used → **BLOCKED**
- **QR-007:** Already-used invitation token is reused → **BLOCKED**
- **QR-008:** Invalid or malformed invitation token is submitted → **BLOCKED**
- **QR-009:** Invitation displays inviter identity and care circle name before acceptance → **SUCCESS**
- **QR-010:** Unauthorized user attempts to generate care-circle invitation → **BLOCKED**

### 12.3.4 Wearable Device Integration Test Coverage

- **WEAR-001:** User connects supported wearable account using mock authorization → **SUCCESS**
- **WEAR-002:** User denies wearable authorization request → **BLOCKED / USER CANCELLED**
- **WEAR-003:** Wearable metric sync receives valid mock data → **SUCCESS**
- **WEAR-004:** Wearable dashboard displays latest synced metric and timestamp → **SUCCESS**
- **WEAR-005:** Wearable metric exceeds configured threshold → **ALERT GENERATED**
- **WEAR-006:** Wearable sync fails due to missing provider access → **ERROR MESSAGE DISPLAYED**
- **WEAR-007:** User revokes wearable consent → **SUCCESS**
- **WEAR-008:** System attempts sync after consent revocation → **BLOCKED**
- **WEAR-009:** Unauthorized caregiver attempts to view patient wearable data → **BLOCKED**
- **WEAR-010:** Stale wearable data is detected after no recent sync → **WARNING**



### 12.3.5 Home Care Document Digitization Test Coverage

- **DOC-001:** User uploads approved home care document type → **SUCCESS**
- **DOC-002:** User uploads unsupported document type → **VALIDATION ERROR**
- **DOC-003:** User submits required document fields with valid input → **SUCCESS**
- **DOC-004:** User submits form with missing required fields → **VALIDATION ERROR**
- **DOC-005:** Document is saved in draft state → **SUCCESS**
- **DOC-006:** Document is submitted for review → **SUCCESS**
- **DOC-007:** Home Care Agency Supervisor approves submitted document → **SUCCESS**
- **DOC-008:** Home Care Agency Supervisor returns document for correction → **SUCCESS**
- **DOC-009:** Unauthorized user attempts to approve document → **BLOCKED**
- **DOC-010:** Submitted document is linked to correct patient/caregiver profile → **SUCCESS**

### 12.3.6 Screening and Assessment Test Coverage

- **ASM-001:** Caregiver starts new screening assessment → **SUCCESS**
- **ASM-002:** Screening assessment saves as draft → **SUCCESS**
- **ASM-003:** User submits screening with required fields complete → **SUCCESS**
- **ASM-004:** User submits screening with missing required fields → **VALIDATION ERROR**
- **ASM-005:** Comprehensive assessment opens required multi-section form → **SUCCESS**
- **ASM-006:** Comprehensive assessment saves progress across sections → **SUCCESS**
- **ASM-007:** User submits comprehensive assessment with all required sections complete → **SUCCESS**
- **ASM-008:** User submits comprehensive assessment with incomplete sections → **VALIDATION ERROR**
- **ASM-009:** Submitted assessment becomes read-only after submission → **SUCCESS**
- **ASM-010:** Supervisor reviews submitted assessment → **SUCCESS**
- **ASM-011:** Unauthorized user attempts to view assessment → **BLOCKED**
- **ASM-012:** High-risk assessment response triggers escalation flag → **SUCCESS**

## 12.4 Test Automation

Automated testing will be used whenever practical to improve test coverage, reduce manual effort, and support regression testing.

Automated testing activities include:

- Backend unit tests for service-layer functionality
- API integration tests
- Form validation tests
- Invitation token validation tests
- EVV workflow validation tests
- Automated regression testing for critical user workflows

Continuous Integration (CI) pipelines will execute automated test suites during development to identify defects early and prevent regressions.

For external dependencies such as EVV aggregators, AWS Bedrock, wearable APIs, and OCR providers, automated tests will utilize mock services and predefined test fixtures until production integrations are approved and available.

The overall objective of test automation is to provide repeatable validation of Team D functionality while supporting rapid development and maintaining software quality throughout the capstone project lifecycle.

## 13. Deployment Strategy

The deployment strategy for Team D backend features is designed to support safe promotion from local development to shared validation and demonstration environments while preserving rollback capability and respecting centralized infrastructure ownership.

### 13.1 Deployment Environments

| Environment | Purpose                                                                                                              | Owner / Status                                           |
|-------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Development | Daily coding, unit testing, service testing, and mock-first workflow validation                                      | Team D / active                                          |
| Test        | Validation before classwide integration, API contract checks, and controlled non-production integration verification | Shared pipeline support with Team D validation           |
| Final Demo  | Stable demonstration of approved workflows and shared hosted release candidate                                       | Shared platform ownership with Team D feature validation |

### 13.2 Infrastructure or Hosting

| Service / Resource                  | Purpose                                                          | Owner                                            | Status                                    |
|-------------------------------------|------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------|
| Spring Boot application runtime     | Hosts Team D backend services, APIs, and orchestration logic     | Shared application hosting model                 | In use                                    |
| PostgreSQL database                 | Stores structured workflow data, states, and metadata            | Shared database platform                         | In use with approved schema scope         |
| Amazon S3 or object-storage pattern | Stores uploaded invoice and document artifacts                   | Shared platform storage                          | Available by design pattern               |
| AWS Bedrock                         | Provides model access for LLM-assisted invoice extraction        | Shared cloud dependency with Team A coordination | Controlled / dependent on approved access |
| CloudWatch-compatible monitoring    | Captures logs, metrics, and operational visibility               | Shared platform service                          | In use                                    |
| KMS and IAM-managed access          | Protects secrets, service access, and infrastructure permissions | Shared platform service                          | In use                                    |
| CI/CD hosted pipeline               | Builds, tests, and promotes approved application artifacts       | Shared platform ownership                        | In use                                    |

### 13.3 CI/CD Pipeline

Team D will deliver code through the shared repository and hosted pipeline rather than a separate deployment path. Changes are developed in the shared codebase, reviewed through the agreed workflow, and validated through build and test stages before promotion. The expected flow is: implement feature changes, run local and module-level tests, submit for review, merge into the approved branch, execute shared build and quality gates, deploy to the test environment, run smoke and workflow validation, and then promote the approved build for final demonstration use. Team A retains ownership of the base CI/CD environment and deployment orchestration, while Team D remains responsible for making its modules buildable, testable, and integration-ready within that shared process.

### 13.4 Rollback Procedure

If a Team D change causes a failed build, broken protected route, invalid state transition, unstable workflow, or unsafe external-service behavior, the response will be to stop forward promotion, identify the offending build or configuration change, and restore the last known stable version through the shared pipeline. Any provider-specific endpoint or feature flag contributing to the defect should be returned to mock, preview, or approved sandbox mode until corrected. After restoration, the team must run smoke checks across login and role protection, EVV core flows, invoice review, invite handling, and any impacted document or assessment APIs before declaring recovery complete. The defect, rollback action, and required remediation should then be documented before another deployment attempt is made.

## 14. Known Issues & Limitations

### 14.1 Technical Limitations

| ID     | Limitation                                                                                                                     | Impact                                                                                                | Mitigation or Next Step                                                                                                 |
|--------|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| LIM-01 | AWS Bedrock credentials not yet provisioned; all Bedrock-dependent features use MockBedrockAdapter in Milestone 2              | Invoice extraction not functional in Milestone 2 against live AI; mock returns predefined test data   | Team A provisions when Anthropic account approved; <code>spring.profiles.active=bedrock</code> switches to live adapter |
| LIM-02 | DeepSeek invoice extraction source file not located; migration targets Bedrock as a net-new implementation                     | No DeepSeek fallback; no regression risk since Bedrock is the target direction                        | No action required; first InvoiceService implementation targets Bedrock API directly                                    |
| LIM-03 | Google Health Connect is an Android only on-device; no equivalent available for iOS in Milestone 2                             | Wearable features unavailable on iOS and web in Milestone 2; graceful message displayed               | Apple HealthKit (iOS) integration targeted for Milestone 3 after Apple developer entitlement confirmed                  |
| LIM-04 | EVV gap analysis is preliminary; final gap list subject to revision                                                            | GAP-01 through GAP-04 may be revised or expanded following Spring 1 code review                       | Code review in Sprint 1 of Milestone 2; TDD to be amended if gaps change                                                |
| LIM-05 | Home Care document forms are design-phase only in Milestone 2; final gap list subject to revision                              | DocumentService accepts uploads and stores in S3 but returns no extracted form data until Milestone 3 | Full OCR pipeline (Tesseract or AWS Textract) targeted for Milestone 3 after technology evaluation                      |
| LIM-06 | Comprehensive assessment JSONB sections (5 per record) may produce records in 5-10 KB in size for verbose clinical assessments | Potential performance impact for large patient cohorts at scale                                       | Pagination required for list queries; JSONB sections excluded from list projections per Section 11                      |

### 14.2 Known Issues

| ID     | Issue                                                                                                                                     | Impact                                                                                                                 | Owner            | Resolution Plan                                                                                                                                                |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KI-001 | AWS Bedrock provisioning blocked pending Anthropic account approval with Team A; cannot test live inference                               | Invoice extraction not functional against live AI in Milestone 2; MockBedrockAdapter returns predefined test data only | Team A           | Team A provisions when Anthropic account approved; <code>spring.profiles.active=bedrock</code> switches to live adapter.                                       |
| KI-002 | EVV codebase identified gaps that may be revised or expanded post-review of implementation                                                | Final gap list subject to change; additional test cases may be required beyond the current 10 EVV cases                | Cristobal Garcia | Full code review by Cristobal Garcia in Sprint 1 of Milestone 2; TDD section 5.1 to be amended if gaps change and corresponding test cases incorporated.       |
| KI-003 | EVV aggregator outreach to 10+ organizations in progress pending Professor response; no confirmed integrations as of Milestone 2 planning | Live aggregator submission cannot be tested; Test cases remain blocked until credentials available                     | Cristobal Garcia | Outreach ongoing via teams-resolution-channel; AggregatorDependency table tracks per-aggregator status (PENDING, CONTACTED, AWAITING_RESPONSE, CREDENTIALS_REC |

|  |  |  |  |                                                                 |
|--|--|--|--|-----------------------------------------------------------------|
|  |  |  |  | EVIED); mock adapter covers all functional tests in Milestone 2 |
|--|--|--|--|-----------------------------------------------------------------|

### 14.3 Exclusions

- Live EVV aggregator API submission
- EHR/FHIR data export or import (future milestone)
- WCAG 2.1 AA formal certification (owned by Team B; post-Milestone 2)
- Multilingual support - Spanish and voice control (owned by Team C in Milestone 2)
- Electronic signature workflows and legal enforceability
- Team A infrastructure: RBAC provisioning, video calls, telemetry/analytics, production deployment pipeline
- Team E features: USPS Informed Delivery, STML/AI agent, call summaries, visit summaries
- HealthKit (iOS) wearable integration (targeted for Milestone 3 after Apple entitlement confirmed)
- Fitbit, Garmin, and other third-party wearable device APIs (future milestones)
- Production AWS Bedrock inference (pending Anthropic account approval via Team A)
- Automated UI/end-to-end test execution (targeted for milestone 3)
- ML model development or custom-tuning of AI models

### 14.4 Future Enhancements

- Home care document digitization - OCR pipeline and field extraction of forms in Milestone 3
- Replace MockBedrockAdapter with live Bedrock implementation when credentials provisioned in Milestone 3
- Apple HealthKit (iOS) wearable integration in Milestone 3 pending Apple entitlement
- Automated end-to-end and UI test suite in Milestone 3
- Fitbit/Garmin wearable device integration as a future enhancement
- EHR/FHIR export for assessment and screening records as a future enhancement
- Predictive health trend analytics using Bedrock for wearable data patterns as a future enhancement
- Electronic signature integration for home care documents as a future enhancement